AN5141
Application note

Results of FMEA on STM32F4 Series microcontrollers

Introduction

This document reports the results the Failure Mode Effect Analysis (FMEA) carried out on microcontrollers of the STM32F4 Series, in compliance with the IEC61508 safety standard.

This document is released to provide, in full detail, the STM32F4 failure modes and the related countermeasures to be implemented for correct failure mitigation.

STM32F4 Safety safety manual (UM1840), available on *www.st.com*, is the reference document, describing in detail the safety concepts.

This document is written in compliance with the IEC 61508 international norm for functional safety of electrical/electronic/programmable electronic safety-related systems. The version used as reference is IEC 61508-1/7:2010.

Section 2 provides basic explanations on the document structure, and *Table 2* reports the actual FMEA results.

Contents

List of tables

1 Terms and abbreviations

Table 1. Terms and abbreviations

Acronym	Definition
CCF	Common cause failure
CM	Continuous mode
COTS	Commercial off-the-shelf
CoU	Conditions of use
CPU	Central processing unit
DC	Diagnostic coverage
DTI	Diagnostic test interval
EUC	Equipment under control
FIT	Failure in time
FMEA	Failure mode effect analysis
FMEDA	Failure mode effect diagnostic analysis
HD	High demand
HFT	Hardware fault tolerance
HW	Hardware
ITRS	International technology roadmap for semiconductors
LD	Low demand
MCU	Microcontroller unit
PFH	Probability of failure per hour
PST	Process safety time
SFF	Safe failure fraction
SIL	Safety integrity level

Notations and acronyms used, if not reported in *Table 1*, are compliant with the functional documentation released for the STM32F4 Series, available on *www.st.com*.

2 FMEA results

Microcontrollers of the STM32F4 Series are based on an Arm®[a] core. The FMEA for these devices is carried out according to ST methodology guidelines for the application of IEC61508 safety standard to microelectronics devices. Detailed information on such methodology is maintained in STM32F4 Safety case as part of List of evidences (refer to UM1840 for details).

The results are detailed in *Table 2*, where

- Failure mode code: is the unique code identifying the failure mode
- Module: indicates the MCU basic function/module where the failure mode belongs
- Submodule: is an additional localization field, used when the FMEA is applied to submodules of the basic functions.
- Potential failure mode
  - Detailed: is the description of the failure mode
  - IEC 61508: provides links between the specific failure mode and general categories as defined in IEC61508-2, Table A.1
- Potential effect(s) on MCU: reports the effect of specified failure mode visible at application level
- Fault model (FM): can be either permanent (P) or transient (T)
- Control: details the mitigation measures[b] available on the MCU hardware
  - Prevention: measures aimed at avoiding the occurrence of malfunctions (e.g. lock mechanisms preventing the access to some programmable registers)
  - Detection: safety mechanisms/measures aimed at detecting and/or correcting hardware malfunctions (e.g. SRAM parity)
- Recommended action(s): lists additional safety mechanisms/measures[b] to be added to achieve the correct mitigation of specified failure mode.

Warning: Microcontrollers of the STM32F4 Series feature different combinations of peripherals (for instance, some of them are not equipped with the USB peripheral). To avoid generating useless information, this document addresses the overall possible peripherals available in the targeted products. End-users must select the peripherals available on their devices, and ignore the FMEA lines related to peripherals/functions not available in their selected MCUs.

a. Arm is a registered trademark of Arm Limited (or its subsidiaries) in the US and/or elsewhere.

b. Acronyms used for safety mechanisms specified here are the same of UM1840. Note that when the mitigation of the failure mode is achieved only with the contribution of a Condition/Assumption of Use, then this field will report its description.

Table 2. FMEA results - STM32F4 Series

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Cortex M4	CORE/ALU	CPU _FM_001	Wrong arithmetic data computation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application data computation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_002				T	-	-	CPU_SM_2: Double computation in application software
		CPU _FM_003	Wrong jump address computation		Wrong application software flow	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_004				T	-	-	CPU_SM_1: Control flow monitoring in application software
	CORE/CTL	CPU _FM_005	Wrong instruction decode and/or execution	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application software flow Wrong application data computation Wrong output signal generation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_006				T	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_2: Double computation in application software
		CPU _FM_007	Wrong data load/store from/to volatile memory		Wrong application software flow Wrong application data computation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_008				T	-	-	RAM_SM_3: Information redundancy for system variables in application software RAM_SM_2: Stack hardening for application software
		CPU _FM_009	Wrong instruction fetch from Flash memory (within application software area)		Wrong application software flow	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_010				T	-		CPU_SM_1: Control flow monitoring in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
Cortex M4	CORE/CTL	CPU _FM_011	Wrong instruction fetch from Flash memory (outside the used area)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application software flow	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_012				T	-		CPU_SM_1: Control flow monitoring in application software FLASH_SM_6: Flash unused area filling code
		CPU _FM_013	Wrong data fetch from Flash memory	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application data computation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_014				T	-	-	FLASH_SM_4: Static data encapsulation
		CPU _FM_015	Unintended entry in Low Power mode		No application software execution	P	-	-	CPU_SM_5: External watchdog
		CPU _FM_016				T	-	-	
		CPU _FM_017	Missed entry in Low Power mode		Wrong application software flow (no low power mode transition)	P	-	-	CPU_SM_1: Control flow monitoring in application software CoU_3: Low power mode state must not be used in safety function(s) implementation
		CPU _FM_018				T	-	-	
	CORE/GPR	CPU _FM_019	Wrong value in general purpose register(s) bit(s)	CPU registers, RAM: DC fault model for data and addresses	Wrong application software flow Wrong application data computation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_020		CPU registers, RAM: change of information caused by soft-errors		T	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_2: Double computation in application software
	CORE/MUL	CPU _FM_021	Wrong data computation in multiplication operation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application data computation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_022				T	-	-	CPU_SM_2: Double computation in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Cortex M4	CORE/GPR	CPU _FM_023	Wrong program counter value (within application software area)	CPU registers, RAM: DC fault model for data and addresses	Wrong application software flow	P	-	-	CPU_SM_1: Control flow monitoring in application software
		CPU _FM_024	Wrong program counter value (outside program used area)			P	-	-	FLASH_SM_6: Flash memory unused area filling code CPU_SM_3: Arm® Cortex® M4 HardFault exceptions
		CPU _FM_025	Program counter frozen		No application software execution	P	-	-	CPU_SM_5: External watchdog
		CPU _FM_026	Wrong program counter value (within application software area)	CPU registers, RAM: change of information caused by soft-errors	Wrong application software flow	T	-	-	CPU_SM_1: Control flow monitoring in application software
		CPU _FM_027	Wrong program counter value (outside program used area)			T	-	-	FLASH_SM_6: Flash memory unused area filling code CPU_SM_3: Arm® Cortex® M4 HardFault exceptions
		CPU _FM_028	Wrong or frozen stack pointer or link register value(s)	CPU registers, RAM: DC fault model for data and addresses	Wrong application software flow Wrong application data computation	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_029	Wrong stack pointer or link register value(s)	CPU registers, RAM: change of information caused by soft-errors		T	-		CPU_SM_4: Stack hardening for application software CPU_SM_1: Control flow monitoring in application software
	CORE/PFU	CPU _FM_030	Wrong pipeline operation on data and/or on instruction code	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application software flow Wrong application data computation	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_031				T	-		CPU_SM_1: Control flow monitoring in application software CPU_SM_2: Double computation in application software
		CPU _FM_032	Missed pipeline operation		Application software executed at reduced speed	P	-	-	CPU_SM_5: External watchdog
		CPU _FM_033			Single instruction executed at reduced speed	T	-	-	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Cortex M4	CORE/PSR	CPU _FM_034	Wrong value in APSR flags (condition flags)	CPU registers, RAM: DC fault model for data and addresses	Wrong application data computation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_035		CPU registers, RAM: change of information caused by soft-errors	Wrong branch decision leading to wrong application software flow	T	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_2: Double computation in application software
		CPU _FM_036	Wrong value of IPSR flags (interrupt priority)	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing exception execution	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
		CPU _FM_037		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		CPU _FM_038	Wrong value in EPSR flags	CPU registers, RAM: DC fault model for data and addresses	Fault exception raise so application program stopped	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	-
		CPU _FM_039		CPU registers, RAM: change of information caused by soft-errors		T	-		-
	CORE/FPU	CPU _FM_040	Wrong floating point computation	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong application data computation	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_041				T	-		CPU_SM_2: Double computation in application software
		CPU _FM_042	Unintended FPU exception raise		Unintended error reporting	P	-		NVIC_SM_1: Expected and unexpected interrupt check by application software CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level
		CPU _FM_043				T	-		
		CPU _FM_044	Wrong value in FPU register(s) bit(s)	CPU registers, RAM: DC fault model for data and addresses	Wrong application data computation	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_045		CPU registers, RAM: change of information caused by soft-errors		T	-	-	CPU_SM_2: Double computation in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
Cortex M4	NVIC/MAIN	CPU _FM_046	No interrupt processing	Interrupt: no or continuous interrupt	Wrong/missing input data for application software	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
		CPU _FM_047			Wrong application software flow	T	-	-	
		CPU _FM_048	Continuous interrupt execution (babbling interrupt)		Application software executed at reduced speed	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software CPU_SM_5: External watchdog
	NVIC/MAIN +REGS	CPU _FM_049	Wrong interrupt priority processing	Interrupt: cross-over of interrupts	Wrong/missing input data for application software Wrong application software flow	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
		CPU _FM_050				T	-	-	
		CPU _FM_051	Execution of a wrong interrupt/exception handler			P	-	-	
		CPU _FM_052				T	-	-	
		CPU _FM_053	Wrong values in NVIC jump table	CPU registers, RAM: DC fault model for data and addresses		P	-	-	NVIC_SM_0: Periodical read-back of configuration registers NVIC_SM_1: Expected and unexpected interrupt check by application software
		CPU _FM_054		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Cortex M4	CORE/MPU	CPU _FM_055	Wrong value in MPU configuration register(s) bit(s)	CPU registers, RAM: DC fault model for data and addresses	MPU misconfiguration leading to unintended memory access violation report	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_056		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		CPU _FM_057		CPU registers, RAM: DC fault model for data and addresses		P	-	-	
		CPU _FM_058		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		CPU _FM_059	Wrong MPU controls leading to unintended memory access violation reporting	CPU coding and execution including flag registers: Wrong coding or wrong execution	Unintended error reporting	P	-	-	CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level
		CPU _FM_060				T	-	-	
		CPU _FM_061			Missing software access error detection	P	-	-	
		CPU _FM_062				T	-	-	
	SYSTICK	CPU _FM_063	Wrong configuration register values	CPU registers, RAM: DC fault model for data and addresses	Application software timing perturbation OS scheduling feature malfunction	P	-	-	CPU_SM_1: Control flow monitoring in application software
		CPU _FM_064		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		CPU _FM_065	No counting	CPU: Coding and execution including flag registers. Wrong coding or wrong execution		P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CPU _FM_066	Too fast or too slow counting			P	-	-	CPU_SM_1: Control flow monitoring in application software
		CPU _FM_067	No clock gating feature when required	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Increase of CPU power consumption	P	-	-	VSUP_SM_3: Internal temperature sensor check
		CPU _FM_068	Unintended gating of clocks		Partial or total loss of CPU functionalities	P	-	-	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core CPU_SM_1: Control flow monitoring in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode – Detailed	Potential failure mode – IEC 61508	Potential effect(s) on the MCU	FM	Control – Prevention	Control – Detection	Recommended action(s)
Cortex M4	BUS INTERFACE	CPU _FM_069	Wrong data and/or addresses in read/write access to an AHB connected slave	Data communication: transmission errors	Execution of illegal instructions Wrong application software execution and flow Wrong application data	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core CPU_SM_5: External watchdog
		CPU _FM_070				T	-		CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CPU _FM_071	Wrong data and/or addresses in read/write access to system RAM	Data communication: transmission errors	Wrong application software execution and flow Wrong application data	P	-	-	RAM_SM_0: Periodical software test for SRAM
		CPU _FM_072				T	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_2: Double computation in application software
		CPU _FM_073	Wrong data and/or addresses in read/write access to system Flash		Wrong application software execution or flow	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	CPU_SM_0: Periodical software test addressing permanent faults in Arm® Cortex® M4 CPU core
		CPU _FM_074				T	-		CPU_SM_1: Control flow monitoring in application software
		CPU _FM_075	Wrong data and/or addresses in read/write access to system peripheral		Application data corruption	P	-	-	BUS_SM_0: Periodical software test for interconnections
		CPU _FM_076				T	-	-	BUS_SM_1: Information redundancy in intra-chip data exchanges
		CPU _FM_077	Inability to access to AHB connected slaves due to arbitration failure	Bus: No or continuous arbitration	No application software execution No access to application data Missing application data	P	-	-	CPU_SM_5: External watchdog CPU_SM_1: Control flow monitoring in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Cortex M4	DEBUG (ETM/TPIU/ITM /DWT/FPB)	CPU _FM_078	Wrong functionality of breakpoint and/or watch point function(s)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No debug feature available	P	-	-	CoU_2: STM32F4 debug features must not be used in safety function(s) implementation
		CPU _FM_079	Unintended activation of breakpoint unit		Wrong application software execution or flow No execution of software application	P	-	-	CPU_SM_5: External watchdog CPU_SM_1: Control flow monitoring in application software
		CPU _FM_080				T	-	-	
		CPU _FM_081	Unintended activation of debugger interface toward CM4 Bus Interface			P	-	-	
		CPU _FM_082				T	-	-	
	WIC	CPU _FM_083	Missing CPU wake-up from suspended mode		No CPU wake-up therefore no application software execution	P	CoU_3: Low power mode state must not be used in safety function(s) implementation	-	CPU_SM_5: External watchdog
		CPU _FM_084				T		-	
	DAP/JTAG	CPU _FM_085	Wrong functionality of Debug Access Port		No debug feature available	P	-	-	CoU_2: STM32F4 debug features must not be used in safety function(s) implementation

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
FLASH	FLASH MACRO	FLASH _FM_001	Single/multiple bit wrong value(s) in main memory block (bit corruption)	Invariable memory: DC fault model for data and addresses	Wrong application software execution Wrong application data	P	-	-	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_002	Single/multiple bit wrong value(s) in information block/BIF (bit corruption)		Wrong MCU settings for HC clock	P	FLASH_SM_3: Option byte write protection	-	CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_003	Single/multiple bit wrong value(s) in information block/SIF (bit corruption)		Diagnostics (RAM parity, sw watchdog) disable	P	-	-	SYSCFG_SM_0: Periodical read-back of configuration registers
	ITF/FPEC	FLASH _FM_004	Unintended Flash memory page erase in main memory block	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No application software execution Wrong application software flow	P	-	FLASH_SM_2: Arm® Cortex® M4 HardFault exceptions	CPU_SM_5: External watchdog FLASH_SM_1: Control flow monitoring in application software
		FLASH _FM_005	Unintended Flash memory page erase in big information block		Wrong MCU settings for HC clock	P	FLASH_SM_3: Option byte write protection	-	CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_006	Missed Flash memory page Erase in main memory block		Wrong application data	P	-	-	FLASH_SM_4: Static data encapsulation
		FLASH _FM_007	Unintended Flash memory mass Erase		No application software execution	P	-	FLASH_SM_2: Arm® Cortex® M4 HardFault exceptions	CPU_SM_5: External watchdog
	ITF/OBL	FLASH _FM_008	Wrong option byte values load on startup/reset		Wrong MCU settings for HC clock	P	FLASH_SM_3: Option byte write protection	FLASH_SM_5: Option byte redundancy with load verification	CPU_SM_1: Control flow monitoring in application software
	ITF/FTL	FLASH _FM_009	Unintended Flash memory test function enable		No application software execution	P	-	-	CPU_SM_5: External watchdog
		FLASH _FM_010				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
FLASH	ITF/RAIN/ ICODE ITF	FLASH _FM_011	Instruction read from wrong address (within the application software area) and/or instruction code read error	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application software execution	P	-	FLASH_SM_2: Arm® Cortex® M4 HardFault exceptions	FLASH_SM_1: Control flow monitoring in application software FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_012				T	-		FLASH_SM_1: Control flow monitoring in application software
		FLASH _FM_013	Instruction read form wrong address (outside the application software area)			P	-		FLASH_SM_6: Flash unused area filling code FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_014				T	-		FLASH_SM_6: Flash unused area filling code
		FLASH _FM_015	Wrong or missing instruction prefetch		Application software executed at reduced speed	P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
	ITF/RAIN/ DCODE ITF	FLASH _FM_016	Data read form wrong address and/or data value read error		Wrong application data	P	-	-	FLASH_SM_4: Static data encapsulation FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_017				T	-	-	FLASH_SM_4: Static data encapsulation
	ITF/RAIN/ CONTROLLER	FLASH _FM_018	Errors in propagating from/to Flash memory data/address		Wrong application software execution Wrong application data	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	FLASH_SM_1: Control flow monitoring in application software FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_019				T	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	FLASH_SM_1: Control flow monitoring in application software
	ITF/RAIN/ ARBITER	FLASH _FM_020	No access to Flash memory allowed or arbitration locked on fixed master	Bus: No or continuous arbitration	No application software execution Wrong application software flow	P	-	-	CPU_SM_5: External watchdog FLASH_SM_1: Control flow monitoring in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
FLASH	ITF/ PREFETCH BUFFER	FLASH _FM_021	Corrupted instruction code or data during prefetch operation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application software execution Wrong application data	P	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_5: External watchdog
		FLASH _FM_022				T	-		FLASH_SM_1: Control flow monitoring in application software
		FLASH _FM_023	Missing prefetch operation or prefetch data/instruction from wrong Flash memory address			P	-	-	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_5: External watchdog
		FLASH _FM_024				T	-	-	FLASH_SM_1: Control flow monitoring in application software
	I-Cache/logic	FLASH _FM_025	Wrong cache write (in data or tag)		Wrong application software execution	P	-	-	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_026				T	-	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_027	Wrong cache hit/read (in data or tag)			P	-	-	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_028				T	-	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_029	Missing cache hit		Application software executed at reduced speed	P	-	-	CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_030			None	T	-	-	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
FLASH	I-Cache/ RAM buffer	FLASH _FM_031	Wrong information in data field	CPU registers, RAM: DC fault model for data and addresses	Wrong application software execution	P	-	-	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_032		CPU registers, RAM: change of information caused by soft-errors		T	-	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_033	Wrong information in tag field	CPU registers, RAM: DC fault model for data and addresses		P	-	-	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_034		CPU registers, RAM: change of information caused by soft-errors		T	-	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_035		CPU registers, RAM: DC fault model for data and addresses	Application software executed at reduced speed Wrong application software execution	P	-	-	FLASH_SM_0: Periodical software test for Flash memory CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_036		CPU registers, RAM: change of information caused by soft-errors	Wrong application software execution	T	-	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions CPU_SM_1: Control flow monitoring in application software
	D-Cache/logic	FLASH _FM_037	Wrong cache write (in data or tag)	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong application data	P	-	-	FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_038				T	-	-	CPU_SM_2: Double computation in application software
		FLASH _FM_039	Wrong cache hit/read (in data or tag)			P	-	-	FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_040				T	-	-	CPU_SM_2: Double computation in application software
		FLASH _FM_041	Missing cache hit		Application software executed at reduced speed	P	-	-	CPU_SM_1: Control flow monitoring in application software
		FLASH _FM_042			None	T	-	-	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
FLASH	D-Cache/ RAM buffer	FLASH _FM_043	Wrong information in data field	CPU registers, RAM: DC fault model for data and addresses	Wrong application data	P	-	-	FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_044		CPU registers, RAM: change of information caused by soft-errors		T	-	-	CPU_SM_2: Double computation in application software
		FLASH _FM_045	Wrong information in tag field	CPU registers, RAM: DC fault model for data and addresses		P	-	-	FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_046		CPU registers, RAM: change of information caused by soft-errors		T	-	-	CPU_SM_2: Double computation in application software
		FLASH _FM_047		CPU registers, RAM: DC fault model for data and addresses		P	-	-	FLASH_SM_0: Periodical software test for Flash memory
		FLASH _FM_048		CPU registers, RAM: change of information caused by soft-errors		T	-	-	CPU_SM_2: Double computation in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
RAM	RAM MACRO	RAM _FM_001	Stuck at 0/1 in single bit on application program variables area	Variable memory: DC fault model for data and addresses	Wrong application data	P	-	RAM_SM_1: Parity bit check	RAM_SM_0: Periodical software test for SRAM RAM_SM_3: Information redundancy for system variables in application software
		RAM _FM_002	Stuck at 0/1 in multiple bits of the same word on application program variables area			P	-		
		RAM _FM_003	Stuck at 0/1 in multiple bits of different words on application program variables area			P	-		
		RAM _FM_004	Bit flip in single bit on application program variables area	Variable memory: Change of information caused by soft-errors		T	-		RAM_SM_3: Information redundancy for system variables in application software
		RAM _FM_005	Bit flip in multiple bits of the same word on application program variables area			T	-		
		RAM _FM_006	Bit flip in multiple bits of different words on application program variables area			T	-		
		RAM _FM_007	Stuck at 0/1 in single bit on stacks area	Variable memory: DC fault model for data and addresses		P	-		RAM_SM_0: Periodical software test for SRAM RAM_SM_2: Stack hardening for application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
RAM	RAM MACRO	RAM _FM_008	Stuck at 0/1 in multiple bits of the same word on stacks area	Variable memory: DC fault model for data and addresses	Wrong application data	P	-	-	RAM_SM_0: Periodical software test for SRAM RAM_SM_2: Stack hardening for application software
		RAM _FM_009	Stuck at 0/1 in multiple bits of different words on stacks area			P	-	-	
		RAM _FM_010	Bit flip in single bit on stacks area	Variable memory: Change of information caused by soft-errors		T	-	-	RAM_SM_2: Stack hardening for application software RAM_SM_3: Information redundancy for system variables in application software
		RAM _FM_011	Bit flip in multiple bits of the same word on stacks area			T	-	-	
		RAM _FM_012	Bit flip in multiple bits of different words on stacks area			T	-	-	
		RAM _FM_013	Stuck at 0/1 in single/multiple bits on RAM area not used by application software	Variable memory: DC fault model for data and addresses		P	-	-	-
		RAM _FM_014	Stuck at 0/1 in single/multiple bits on RAM area not used by application software	Variable memory: Change of information caused by soft-errors	None	T	-	-	-
		RAM _FM_015	Single bit wrong value due to coupling with another memory cell	Variable memory: Dynamic cross-over for memory cells	Wrong application data	P	-	-	RAM_SM_2: Stack hardening for application software RAM_SM_3: Information redundancy for system variables in application software
		RAM _FM_016	Multiple bits wrong values due to coupling with another memory cell			P	-	-	
		RAM _FM_017	Read/write on wrong memory address when performing a program variable access	Variable memory: DC fault model for data and addresses		P	-	-	
		RAM _FM_018	Read/write on wrong memory address when performing a stack access operation			P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
RAM	INTERFACE	RAM _FM_019	Data/address wrong values during SRAM accesses by application software by AHB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong data read from SRAM Wrong data write on SRAM	P	-	-	RAM_SM_0: Periodical software test for SRAM
		RAM _FM_020				T	-	-	RAM_SM_3: Information redundancy for system variables in application software CPU_SM_1: Control flow monitoring in application software
		RAM _FM_021	Timeout or no arbitration in SRAM AHB slave interface	Bus: No or continuous arbitration	No application software execution	P	-	-	CPU_SM_5: External watchdog
	CONTROLLER	RAM _FM_022	Wrong address management during read/write access to SRAM	CPU, Address calculation: DC fault model	Wrong application data Wrong application software flow	P	-	-	RAM_SM_0: Periodical software test for SRAM
		RAM _FM_023		CPU, Address calculation: Change of addresses caused by soft-errors		T	-	-	RAM_SM_3: Information redundancy for system variables in application software CPU_SM_1: Control flow monitoring in application software
		RAM _FM_024	Wrong data value during read/write access to SRAM	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application data	P	-	-	RAM_SM_0: Periodical software test for SRAM
		RAM _FM_025				T	-	-	RAM_SM_3: Information redundancy for system variables in application software CPU_SM_1: Control flow monitoring in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
AHB2APB	-	BUS _FM_001	Address wrong values in translation between AHB and APB	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Missing access to peripherals	P	-	-	BUS_SM_2: Periodical read-back of configuration registers
		BUS _FM_002			Missing single data transaction to peripherals Wrong data to/from peripherals	T	-	-	BUS_SM_1: Information redundancy in intra-chip data exchanges
		BUS _FM_003	Data wrong values in translation between AHB and APB		Wrong data from/to peripherals	P	-	-	BUS_SM_2: Periodical read-back of configuration registers
		BUS _FM_004				T	-	-	BUS_SM_1: Information redundancy in intra-chip data exchanges
		BUS _FM_005	Timeout or no arbitration in translation between AHB and APB	Bus: No or continuous arbitration	Missing access to peripherals	P	-	-	BUS_SM_2: Periodical read-back of configuration registers
Bus matrix	-	BUS _FM_006	Data/address wrong values connecting Master 0/1/2 (CPU) with Slave 2/3 (SRAM)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application data	P	-	-	RAM_SM_0: Periodical software test for SRAM
		BUS _FM_007				T	-	-	RAM_SM_3: Information redundancy for system variables in application software RAM_SM_2: Stack hardening for application software
		BUS _FM_008	Timeout or no arbitration for connection between Master 0/1/2 (CPU) and Slave 2/3 (SRAM)	Bus: No or continuous arbitration	No application software execution	P	-	-	CPU_SM_5: External watchdog
		BUS _FM_009	Data/address wrong values connecting Master 0/1/2 (CPU) with Slave 2/3 (SRAM)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application software execution	P	-	-	CPU_SM_3: Arm® Cortex® M4 HardFault exceptions CPU_SM_1: Control flow monitoring in application software
		BUS _FM_010			Wrong application data	T	-	-	
		BUS _FM_011	Timeout or no arbitration for connection between Master 0/1/2 (CPU) and Slave 2/3 (SRAM)	Bus: No or continuous arbitration	No application software execution	P	-	-	CPU_SM_5: External watchdog

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Bus matrix	-	BUS _FM_012	Timeout or no arbitration for connection between Master 0/1 (CPU) and Slave 0/1 (Flash memory)	Bus: No or continuous arbitration	Wrong data from/to peripherals Missing access to peripherals	P	-	-	BUS_SM_2: Periodical read-back of configuration registers BUS_SM_1: Information redundancy in intra-chip data exchanges
Bus matrix	-	BUS _FM_013	Data/address wrong values connecting Master 2 (CPU) with Slave 4/5 (AHB/AHB2APB)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong data from/to peripherals Missing access to peripherals	T	-	-	BUS_SM_1: Information redundancy in intra-chip data exchanges
Bus matrix	-	BUS _FM_014	Data/address wrong values connecting Master 2 (CPU) with Slave 4/5 (AHB/AHB2APB)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong data from/to peripherals Missing access to peripherals	P	-	-	BUS_SM_2: Periodical read-back of configuration registers
Bus matrix	-	BUS _FM_015	Timeout or no arbitration for connection between Master 2(CPU) and Slave 4/5 (AHB/AHN2APB)	Bus: No or continuous arbitration	Wrong data from/to peripherals Missing access to peripherals	P	-	-	BUS_SM_2: Periodical read-back of configuration registers
Bus matrix	-	BUS _FM_016	Data/address wrong values connecting 0/1/2 (CPU) with Slave 6 (FMC/QUADSPI)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application software execution	P	-	-	FLASH_SM_1: Control flow monitoring in application software FLASH_SM_4: Static data encapsulation
Bus matrix	-	BUS _FM_017	Data/address wrong values connecting 0/1/2 (CPU) with Slave 6 (FMC/QUADSPI)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application data	T	-	-	FLASH_SM_1: Control flow monitoring in application software FLASH_SM_4: Static data encapsulation
Bus matrix	-	BUS _FM_018	Timeout or no arbitration for connection between Master 0/1/2 (CPU) and Slave 6 (FMC/QUADSPI)	Bus: No or continuous arbitration	No application software execution	P	-	-	FLASH_SM_1: Control flow monitoring in application software
Bus matrix	-	BUS _FM_019	Data/address wrong values connecting Master 3/4 (DMA) with Slave 2/3 (SRAM)	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong peripherals data transaction by DMA	P	-	-	DMA_SM_1: Information redundancy on data packet transferred via DMA DMA_SM_0: Periodical read-back of configuration registers
Bus matrix	-	BUS _FM_020	Data/address wrong values connecting Master 3/4 (DMA) with Slave 2/3 (SRAM)	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong peripherals data transaction by DMA	T	-	-	DMA_SM_1: Information redundancy on data packet transferred via DMA
Bus matrix	-	BUS _FM_021	Timeout or no arbitration for connection between Master 3/4 (DMA) and Slave 2/3 (SRAM)	Bus: No or continuous arbitration	No peripherals data transaction by DMA	P	-	-	DMA_SM_1: Information redundancy on data packet transferred via DMA DMA_SM_0: Periodical read-back of configuration registers

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Bus matrix	-	BUS _FM_022	Data/address wrong values connecting Master 3/4 (DMA) with Slave 1 (Flash D-code)	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong CRC computation over Flash memory sections	P	-	-	DMA_SM_1: Information redundancy on data packet transferred via DMA DMA_SM_4: DMA transaction awareness checks
		BUS _FM_023			Wrong data read from Flash memory by DMA	T	-	-	
		BUS _FM_024	Timeout or no arbitration for connection between Master 3/4(DMA) and Slave 1(Flash D-code)	Bus: No or continuous arbitration		P	-	-	DMA_SM_4: DMA transaction awareness checks
		BUS _FM_025	Data/address wrong values connecting Master 3/4(DMA) with Slave 4/5 (AHB/AHB2APB)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong peripherals data transaction by DMA	P	-	-	BUS_SM_1: Information redundancy in intra-chip data exchanges
		BUS _FM_026				T	-	-	
		BUS _FM_027	Timeout or no arbitration for connection between Master 3/4(DMA) and Slave 4/5 (AHB/AHN2APB)	Bus: No or continuous arbitration	No peripherals data transaction by DMA	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
		BUS _FM_028	Data/address wrong values connecting Master 3/4 (DMA) with Slave 4/5 (AHB/AHN2APB)	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong application data	P	-	-	FLASH_SM_4: Static data encapsulation
		BUS _FM_029				T	-	-	
		BUS _FM_030	Timeout or no arbitration for connection between Master 3/4 (DMA) and Slave 6 (FMC/QUADSPI)	Bus: No or continuous arbitration	Wrong or missing application data	P	-	-	FLASH_SM_4: Static data encapsulation CPU_SM_1: Control flow monitoring in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
GPIO	INTERFACE	GPIO _FM_001	Data/address wrong values during GPIO module accesses by application software	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong GPIO output signals status Wrong GPIO input data	P	-	-	GPIO_SM_0: Periodical read-back of configuration registers
		GPIO _FM_002				T	-	-	
		GPIO _FM_003	Timeout or no arbitration in GPIO interface	Bus: No or continuous arbitration		P	-	-	
		GPIO _FM_004	GPIO (I/O configuration options) configuration registers wrong values	CPU registers, RAM: DC fault model for data and addresses		P	-	-	
		GPIO _FM_005				T	-	-	
		GPIO _FM_006	GPIO (alternate functions options) configuration registers wrong values	CPU registers, RAM: DC fault model for data and addresses	Wrong behavior of externally connected peripheral (communication/ADC/DAC)	P	-	-	
		GPIO _FM_007				T	-	-	
		GPIO _FM_008	GPIO interrupt configuration registers wrong values	CPU registers, RAM: DC fault model for data and addresses	Missing or unexpected interrupts events triggering	P	-	-	GPIO_SM_0: Periodical read-back of configuration registers NVIC_SM_1: Expected and unexpected interrupt check by application software
		GPIO _FM_009				T	-	-	
	IOPORT	GPIO _FM_010	GPIO input line stuck at 0/1 or open/HiZ	Digital I/O: DC fault model	Wrong GPIO input data	P	-	-	GPIO_SM_1: 1oo2 for input GPIO lines
		GPIO _FM_011	GPIO input line bridged to another GPIO line			P	-	-	
		GPIO _FM_012	GPIO output line stuck at 0/1 or open/HiZ		Wrong GPIO output signals status	P	-	-	GPIO_SM_2: Loopback scheme for output GPIO lines
		GPIO _FM_013	GPIO output line bridged to another GPIO line			P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
GPIO	AFOCTL	GPIO _FM_014	GPIO line not connected to SoC outside	CPU registers, RAM: DC fault model for data and addresses	Wrong GPIO input or output signals status	P	-	-	GPIO_SM_1: 1oo2 for input GPIO lines GPIO_SM_2: Loopback scheme for output GPIO lines
		GPIO _FM_015				T	-	-	
		GPIO _FM_016	Alternate function signal not connected to SoC outside		Wrong alternate function signal value (input/output)	P	-	-	CoU_4: End User must implement the required combination of safety mechanism/CoUs for each STM32 peripheral used in safety function(s) implementation
		GPIO _FM_017				T	-	-	
DMA1/DMA2/DMA2D	AHB SLAVE ITF	DMA _FM_001	Data/address wrong values during DMA accesses by application software by AHB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong configuration for DMA functions	P	-	-	DMA_SM_0: Periodical read-back of configuration registers
		DMA _FM_002				T	-	-	
		DMA _FM_003	Timeout or no arbitration in DMA interface	Bus: No or continuous arbitration	No configuration for DMA functions	P	-	-	
	REGISTERS	DMA _FM_004	DMA peripherals and memory address registers wrong values	CPU registers, RAM: DC fault model for data and addresses	DMA transaction (read and/or write) to/from wrong addresses and/or to/from wrong peripheral/target	P	-	-	DMA_SM_2: Information redundancy including sender/receiver identifier on data packet transferred via DMA
		DMA _FM_005				T	-	-	
		DMA _FM_006	DMA configuration registers wrong values	CPU registers, RAM: DC fault model for data and addresses	No or wrong configuration for DMA functions Unintended enable of DMA functions Missing or wrong execution of DMA functions Wrong channel priority management	P	-	-	DMA_SM_4: DMA transaction awareness checks DMA_SM_3: Periodical software test for DMA
		DMA _FM_007		CPU registers, RAM: change of information caused by soft-errors		T	-	-	DMA_SM_4: DMA transaction awareness checks
		DMA _FM_008	DMA interrupt configuration registers wrong values	CPU registers, RAM: DC fault model for data and addresses	Unintended interrupt enable of DMA functions Missing or wrong interrupt management of DMA functions	P	-	-	DMA_SM_4: DMA transaction awareness checks NVIC_SM_1: Expected and unexpected interrupt check by application software
		DMA _FM_009		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
DMA1/DMA2/DMA2D	CHANNELS	DMA _FM_010	Wrong data values during data transfer operation by the channel	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong data sent to peripherals Wrong data read from peripherals Wrong peripheral data copied to SRAM	P	-	-	DMA_SM_1: Information redundancy on data packet transferred via DMA
		DMA _FM_011				T	-	-	
		DMA _FM_012	Wrong peripheral address values during data transfer operation by the channel		Wrong data read from peripherals No data sent to peripherals	P	-	-	DMA_SM_2: Information redundancy including sender/receiver identifier on data packet transferred via DMA DMA_SM_4: DMA transaction awareness checks
		DMA _FM_013				T	-	-	
		DMA _FM_014	Wrong memory address values during data transfer operation by the channel	CPU, Address calculation: DC fault model	SRAM variables and/or stack overwrite Missing data from peripherals	P	-	-	RAM_SM_3: Information redundancy for system variables in application software DMA_SM_4: DMA transaction awareness checks
		DMA _FM_015		CPU, Address calculation: Change of addresses caused by soft-errors		T	-	-	
		DMA _FM_016	Wrong address recomputation in pointer handling by the channel	CPU, Address calculation: DC fault model	SRAM variables and/or stack overwrite Missing data from peripherals Corruption of data from peripheral Missing or incorrect order of data sent to peripherals	P	-	-	RAM_SM_3: Information redundancy for system variables in application software DMA_SM_2: Information redundancy including sender/receiver identifier on data packet transferred via DMA
		DMA _FM_017		CPU, Address calculation: Change of addresses caused by soft-errors		T	-	-	
		DMA _FM_018	Missing interrupt generation by channel at the end of transaction	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Missing peripherals data usage by application software	P	-	-	DMA_SM_4: DMA transaction awareness checks
		DMA _FM_019				T	-	-	
		DMA _FM_020	Continuous or babbling interrupt generation by the channel		Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode – Detailed	Potential failure mode – IEC 61508	Potential effect(s) on the MCU	FM	Control – Prevention	Control – Detection	Recommended action(s)
DMA1/DMA2/DMA2D	ARBITER	DMA _FM_021	Wrong priority management between channels	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Loss of DMA transaction leading to potential loss of data from peripherals	P	-	-	DMA_SM_4: DMA transaction awareness checks
		DMA _FM_022			Slightly appreciable delay of DMA transactions	T	-	-	
		DMA _FM_023	Masking of channel transaction request		Loss of data from peripherals	P	-	-	
		DMA _FM_024				T	-	-	
	AHB MASTER ITF	DMA _FM_025	Data/address wrong values during DMA accesses to slaves by AHB bus		Data corruption on every DMA transaction	P	-	-	DMA_SM_1: Information redundancy on data packet transferred via DMA DMA_SM_4: DMA transaction awareness checks
		DMA _FM_026			Data corruption or missing data affecting DMA transaction	T	-	-	
		DMA _FM_027	Timeout or no arbitration in DMA interface	Bus: No or continuous arbitration	No DMA transactions possible	P	-	-	DMA_SM_4: DMA transaction awareness checks
USART/UART/LPUART	APB ITF	USART _FM_001	Data/address wrong values during USART accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to USART peripheral by application software and/or DMA	P	-	-	USART_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		USART _FM_002	Data/address wrong values during USART accesses by application software and/or DMA by APB bus			T	-	-	USART_SM_2: Information redundancy techniques on messages DMA_SM_4: DMA transaction awareness checks
		USART _FM_003		Bus: No or continuous arbitration	No access to USART peripheral by application software or DMA APB bus lock	P	-	-	USART_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	USART _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No USART communication	P	-	-	USART_SM_0: Periodical read-back of configuration registers
		USART _FM_005				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
USART/UART/LPUART	BAUD GENERATOR	USART _FM_006	Wrong or missing clock generation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No USART communication	P	-	-	USART_SM_3: Information redundancy techniques on messages, including End to End safing
	INTERRUPT CTRL	USART _FM_007	Missing interrupt generation on receive/transmit data		No USART data processing by application software	P	-	-	USART_SM_3: Information redundancy techniques on messages, including End to End safing NVIC_SM_1: Expected and unexpected interrupt check by application software
		USART _FM_008			Missing transmission of USART data	T	-	-	
		USART _FM_009	Continuous or unintended USART interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software
	OUTPUTS	USART _FM_010	TX and/or RX signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing USART data received Wrong or missing USART data transmitted	P	-	USART_SM_1: Protocol error signals	USART_SM_3: Information redundancy techniques on messages, including End to End safing
		USART _FM_011	nRTS and/or rCTS and/or SCLK signals wrong values, open or HiZ		Wrong or missing USART data received/transmitted in synchronous communication links	P	-		
		USART _FM_012	IrDA_OUT, IrDA_IN signals wrong value, open or HiZ		Wrong or missing data received/transmitted in IRDA communication link	P	-		
	CONTROL	USART _FM_013	Wrong data processing in receive/transmit processing control units	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong or missing USART data received	P	-	-	
		USART _FM_014			Wrong or missing USART data transmitted	T	-	-	
		USART _FM_015	Missed USART wakeup on transmission/receive request	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Missing USART data received Missing USART data transmitted	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
USART/UART/LPUART	-	USART _FM_016	USART communication affected by transmission errors	Data communication: transmission errors	Wrong USART data received Wrong USART data transmitted	P	-	-	USART_SM_2: Information redundancy techniques on messages
		USART _FM_017	USART communication affected by transmission repetitions	Data communication: transmission repetitions		P	-	-	
		USART _FM_018	USART communication affected by transmission deletion	Data communication: transmission deletion	Missing USART data received Missing USART data transmitted	P	-	-	
		USART _FM_019	USART communication affected by transmission resequencing	Data communication: transmission resequencing	Not possible (no internal buffer in USART)	P	-	-	
		USART _FM_020	USART communication affected by transmission corruption	Data communication: transmission corruption	Wrong USART data received Wrong USART data transmitted	P	-	-	USART_SM_2: Information redundancy techniques on messages
		USART _FM_021	USART communication affected by transmission delay	Data communication: transmission delay	Wrong or missing USART data received Wrong or missing USART data transmitted	P	-	-	USART_SM_2: Information redundancy techniques on messages
		USART _FM_022	USART communication affected by transmission masquerading	Data communication: transmission masquerading	Not possible (point/point connections are typical schemes for USART/SDA)	P	-	-	USART_SM_2: Information redundancy techniques on messages

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
SPI	APB ITF	SPI _FM_001	Data/address wrong values during SPI accesses by application software and/or DMA by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to SPI peripheral by application software and/or DMA	P	-	-	SPI_SM_0: Periodical read-back of configuration registers
		SPI _FM_002				T	-	-	SPI_SM_2: Information redundancy techniques on messages
		SPI _FM_003	Timeout or no arbitration in SPI APB interface	Bus: No or continuous arbitration	No access to SPI peripheral by application software or DMA APB bus lock	P	-	-	SPI_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	SPI _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No SPI communication	P	-	-	SPI_SM_0: Periodical read-back of configuration registers
		SPI _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	BAUD GENERATOR	SPI _FM_006	Wrong or missing clock generation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution		P	-	-	SPI_SM_4: Information redundancy techniques on messages, including End to End safing
	INTERRUPT CTRL	SPI _FM_007	Missing interrupt generation on receive/transmit data	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No SPI data processing by application software Missing transmission of SPI data	P	-	-	SPI_SM_4: Information redundancy techniques on messages, including End to End safing NVIC_SM_1: Expected and unexpected interrupt check by application software
		SPI _FM_008			No SPI data processing by application software Missing transmission of SPI data SPI receive buffer overrun	T	-	-	
		SPI _FM_009	Continuous or unintended SPI interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
SPI	OUTPUTS	SPI _FM_010	MISO and/or MOSI signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing SPI data received Wrong or missing SPI data transmitted	P	-	SPI_SM_1: Protocol error signals	SPI_SM_4: Information redundancy techniques on messages, including End to End safing
		SPI _FM_011	SCLK signals wrong value, open or HiZ		Missing SPI data communication	P	-		
		SPI _FM_012	NSS signals wrong value, open or HiZ		Wrong or missing data received/transmitted in multimaster connection	P	-		
	CONTROL	SPI _FM_013	Wrong data processing in receive/transmit processing control units	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong or missing SPI data received	P	-	SPI_SM_3: CRC insertion in SPI message	SPI_SM_4: Information redundancy techniques on messages, including End to End safing
		SPI _FM_014			Wrong or missing SPI data transmitted	T	-		
		SPI _FM_015	Wrong NSS management		Wrong or missing data received/transmitted in multimaster network	P	-	-	
		SPI _FM_016				T	-	-	SPI_SM_2: Information redundancy techniques on messages
	FIFOS	SPI _FM_017	Single or multiple bit errors in Rx/Tx FIFO cells	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing SPI data received	P	-	SPI_SM_3: CRC insertion in SPI message	-
		SPI _FM_018			Wrong or missing SPI data transmitted	T	-		-
	CRC CONTROLLER	SPI _FM_019	Wrong CRC computation for outgoing message	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong CRC field in outgoing message	P	-	SPI_SM_3: CRC insertion in SPI message	-
		SPI _FM_020				P	-		-
		SPI _FM_021	Wrong CRC computation for incoming message		CRC mismatch in well-received message	P	-		-
		SPI _FM_022				P	-		-
		SPI _FM_023	CRC error reporting disable		CRC error protection disable	P	-	-	SPI_SM_5: Software test for CRC error detection on SPI module
		SPI _FM_024	Unintended CRC error reporting		CRC error propagated so system will switch in safe state	P	-	SPI_SM_3: CRC insertion in SPI message	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode – Detailed	Potential failure mode – IEC 61508	Potential effect(s) on the MCU	FM	Control – Prevention	Control – Detection	Recommended action(s)
SPI	-	SPI _FM_025	SPI communication affected by transmission errors	Data communication: transmission errors	Wrong SPI data received Wrong SPI data transmitted	P	-	SPI_SM_3: CRC insertion in SPI message	-
		SPI _FM_026	SPI communication affected by transmission repetitions	Data communication: transmission repetitions		P	-		SPI_SM_4: Information redundancy techniques on messages, including End to End safing
		SPI _FM_027	SPI communication affected by transmission deletion	Data communication: transmission deletion	Missing SPI data received Missing SPI data transmitted	P	-		-
		SPI _FM_028	SPI communication affected by transmission resequencing	Data communication: transmission resequencing	SPI data received in wrong sequence SPI data transmitted in wrong sequence	P	-		-
		SPI _FM_029	SPI communication affected by transmission corruption	Data communication: transmission corruption	Wrong SPI data received Wrong SPI data transmitted	P	-		-
		SPI _FM_030	SPI communication affected by transmission delay	Data communication: transmission delay	Wrong or missing SPI data received Wrong or missing SPI data transmitted	P	-	SPI_SM_1: Protocol error signals	SPI_SM_4: Information redundancy techniques on messages, including End to End safing
		SPI _FM_031	SPI communication affected by transmission masquerading	Data communication: transmission masquerading	Wrong SPI message received in multimaster network	P	-	-	-
I2C	APB ITF	IIC _FM_001	Data/address wrong values during I2C accesses by application software and/or DMA by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to I2C peripheral by application software and/or DMA	P	-	-	IIC_SM_0: Periodical read-back of configuration registers
		IIC _FM_002				T	-	-	IIC_SM_4: Information redundancy techniques on messages, including End to End safing DMA_SM_4: DMA transaction awareness checks
		IIC _FM_003	Timeout or no arbitration in I2C APB interface	Bus: No or continuous arbitration	No access to I2C peripheral by application software or DMA APB bus lock	P	-	-	IIC_SM_0: Periodical read-back of configuration registers

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode – Detailed	Potential failure mode – IEC 61508	Potential effect(s) on the MCU	FM	Control – Prevention	Control – Detection	Recommended action(s)
I2C	REGISTERS	IIC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No I2C communication	P	-	-	IIC_SM_0: Periodical read-back of configuration registers
		IIC _FM_005				T	-	-	
	BAUD GENERATOR	IIC _FM_006	Wrong or missing clock generation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution		P	-	-	IIC_SM_4: Information redundancy techniques on messages, including End to End safing
	INTERRUPT CTRL	IIC _FM_007	Missing interrupt generation on receive/transmit data	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No I2C data processing by application software Missing transmission of I2C data	P	-	-	IIC_SM_4: Information redundancy techniques on messages, including End to End safing NVIC_SM_1: Expected and unexpected interrupt check by application software
		IIC _FM_008			No I2C data processing by application software Missing transmission of I2C data I2C receive buffer overrun	T	-	-	
		IIC _FM_009	Continuous or unintended I2C interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	
	OUTPUTS	IIC _FM_010	SDA and/or SCL signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing I2C data received Wrong or missing I2C data transmitted	P	-	IIC_SM_1: Protocol error signals	IIC_SM_4: Information redundancy techniques on messages, including End to End safing
		IIC _FM_011	SMBA signals wrong value, open or HiZ		Wrong or missing data received/transmitted in multislave network	P	-		

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
I2C	DATA/CLOCK CONTROL	IIC _FM_012	Wrong data processing in receive/transmit processing control units	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong or missing I2C data received in SMBUS communication	P	-	-	IIC_SM_3: CRC payload in SMBUS communications
		IIC _FM_013				T	-	-	
		IIC _FM_014			Wrong or missing I2C data transmitted in SMBUS communication	P	-	-	IIC_SM_4: Information redundancy techniques on messages, including End to End safing
		IIC _FM_015				T	-	-	
	WAKEUP CONTROL	IIC _FM_016	Missing wakeup on address match		Wrong or missing I2C data received	P	-	-	IIC_SM_4: Information redundancy techniques on messages, including End to End safing DMA_SM_4: DMA transaction awareness checks
		IIC _FM_017				T	-	-	
	SMBUS CONTROLLER	IIC _FM_018	Wrong CRC computation for outgoing SMBUS message	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Wrong CRC field in SMBUS outgoing message	P	-	IIC_SM_3: CRC payload in SMBUS communications	-
		IIC _FM_019				T	-		-
		IIC _FM_020			CRC mismatch in well-received SMBUS message	P	-		-
		IIC _FM_021				T	-		-
		IIC _FM_022	CRC error reporting disable	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	CRC error protection disable	P	-	-	IIC_SM_5: Software test for CRC error detection on I2C module
		IIC _FM_023	Unintended CRC error reporting	-	CRC error propagated so system will switch in safe state	P	-	IIC_SM_3: CRC payload in SMBUS communications	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
I2C	-	IIC _FM_024	I2C communication affected by transmission errors	Data communication: transmission errors	Wrong I2C data received Wrong I2C data transmitted	P	-	-	IIC_SM_2: Information redundancy techniques on messages
		IIC _FM_025	I2C communication affected by transmission repetitions	Data communication: transmission repetitions		P	-	-	IIC_SM_4: Information redundancy techniques on messages, including End to End safing
		IIC _FM_026	I2C communication affected by transmission deletion	Data communication: transmission deletion		P	-	-	
		IIC _FM_027	I2C communication affected by transmission resequencing	Data communication: transmission resequencing	I2C data received in wrong sequence I2C data transmitted in wrong sequence	P	-	-	IIC_SM_2: Information redundancy techniques on messages
		IIC _FM_028	I2C communication affected by transmission corruption	Data communication: transmission corruption	Wrong I2C data received Wrong I2C data transmitted	P	-	-	
		IIC _FM_029	I2C communication affected by transmission delay	Data communication: transmission delay	Wrong or missing I2C data received Wrong or missing I2C data transmitted	P	-	-	IIC_SM_4: Information redundancy techniques on messages, including End to End safing
		IIC _FM_030	I2C communication affected by transmission masquerading	Data communication: transmission masquerading	Wrong I2C message received in multimaster network	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
USB	APB ITF	USB _FM_001	Data/address wrong values during USB accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to USB peripheral by application software and/or DMA	P	-	-	USB_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		USB _FM_002	Data/address wrong values during USB accesses by application software and/or DMA by APB bus			T	-	-	USB_SM_3: Information redundancy techniques on messages, including End to End safing
		USB _FM_003		Bus: No or continuous arbitration	No access to USB peripheral by application software or DMA APB bus lock	P	-	-	USB_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	CONFIG REGISTERS	USB _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No or wrong USB communication	P	-	-	USB_SM_0: Periodical read-back of configuration registers
		USB _FM_005				T	-	-	
	ENDPOINT REGISTERS	USB _FM_006	Wrong endpoint register(s) value		No USB communication with the counterpart	P	-	-	
		USB _FM_007				T	-	-	
	INTERRUPT CTRL	USB _FM_008	Missing interrupt generation on receive/transmit data	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No USB data processing by application software	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software USB_SM_3: Information redundancy techniques on messages, including End to End safing
		USB _FM_009			Missing transmission of USB data	T	-	-	
		USB _FM_010	Continuous or unintended USB interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	
	SERIAL INTERFACE ENGINE	USB _FM_011	Wrong CRC computation for incoming/outgoing USB message	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	CRC mismatch in well-received USB message	P	-	USB_SM_2: Information redundancy techniques on messages	-
		USB _FM_012				P	-		-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
USB	SERIAL INTERFACE ENGINE	USB _FM_013	CRC error reporting disable	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	CRC error protection disable	P	-	-	USB_SM_5: Software test for CRC error detection on USB module
		USB _FM_014	Unintended CRC error reporting		CRC error propagated so system will switch in safe state	P	-	USB_SM_2: Information redundancy techniques on messages	-
		USB _FM_015	Wrong data processing in receive/transmit processing control		Wrong or missing USB data received	P	-		-
		USB _FM_016			Wrong or missing USB data transmitted	T	-		-
		USB _FM_017	Wrong endpoint selection or management		No USB communication with the counterpart	P	-	-	USB_SM_3: Information redundancy techniques on messages, including End to End safing
		USB _FM_018				T	-	-	
		USB _FM_019	Wrong encoding/generation of USB PHY signals		Wrong or missing USB data transmitted/received	P	-	USB_SM_1: Protocol error signals (including signals encoding)	
		USB _FM_020				T	-		
	TIMER	USB _FM_021	Missed USB wakeup on transmission/receive request		Missing USB data received Missing USB data transmitted	P	-	-	
	PACKET BUFFERS	USB _FM_022	Wrong packet buffer management in buffer interface during transmission and/or reception		Wrong or missing USB data transmitted/received	P	-	-	
		USB _FM_023			Incorrect data order in USB message	T	-	-	
		USB _FM_024	Wrong value(s) in packet buffers		Wrong or missing USB data transmitted/received	P	-	-	
		USB _FM_025				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
USB	TRANSCEIVER	USB _FM_026	Wrong generation of differential signals output	Analog I/O: DC fault model	Wrong or missing USB data transmitted No USB communication	P	-	USB_SM_1: Protocol error signals (including signals encoding)	USB_SM_3: Information redundancy techniques on messages, including End to End safing
		USB _FM_027	Wrong generation of differential signals input			P	-		
		USB _FM_028	Wrong pullups connection or wrong pullups values		Wrong USB data received Wrong USB data transmitted	P	-		-
	BCD	USB _FM_029	Failures of battery charging detection circuit		Wrong detection of battery charging related status	P	-		USB_SM_3: Information redundancy techniques on messages, including End to End safing
	-	USB _FM_030	USB communication affected by transmission errors	Data communication: transmission errors	Wrong USB data received Wrong USB data transmitted	P	-	USB_SM_2: Information redundancy techniques on messages	-
		USB _FM_031	USB communication affected by transmission repetitions	Data communication: transmission repetitions		P	-		USB_SM_3: Information redundancy techniques on messages, including End to End safing
		USB _FM_032	USB communication affected by transmission deletion	Data communication: transmission deletion	Missing USB data received Missing USB data transmitted	P	-		
		USB _FM_033	USB communication affected by transmission resequencing	Data communication: transmission resequencing	USB data packet received or transmitted with wrong data sequence	P	-	-	
		USB _FM_034	USB communication affected by transmission corruption	Data communication: transmission corruption	Wrong USB data received Wrong USB data transmitted	P	-	USB_SM_2: Information redundancy techniques on messages	-
		USB _FM_035	USB communication affected by transmission delay	Data communication: transmission delay	Wrong or missing USB data received Wrong or missing USB data transmitted	P	-	-	USB_SM_3: Information redundancy techniques on messages, including End to End safing
		USB _FM_036	USB communication affected by transmission masquerading	Data communication: transmission masquerading	MCU will receive an unexpected USB message to be not consumed by the MCU itself	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
General purpose / advanced timers	APB ITF	ATIM _FM_001	Data/address wrong values during TIMER accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to TIMER registers by application software	P	ATIM_SM_4: Timer lock configuration protection	-	ATIM_SM_0: Periodical read-back of configuration registers
		ATIM _FM_002				T		-	
	CONFIG REGISTERS	ATIM _FM_003	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No or wrong counting function	P	-	-	ATIM_SM_0: Periodical read-back of configuration registers NVIC_SM_1: Expected and unexpected interrupt check by application software
		ATIM _FM_004		CPU registers, RAM: change of information caused by soft-errors	No or wrong interrupt generation on timer expiration	T	-	-	
		ATIM _FM_005		CPU registers, RAM: DC fault model for data and addresses	No or wrong frequency/duty PWM signal generation	P	-	-	ATIM_SM_0: Periodical read-back of configuration registers
		ATIM _FM_006		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		ATIM _FM_007		CPU registers, RAM: DC fault model for data and addresses	No or wrong input capture function	P	-	-	
		ATIM _FM_008		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	PRESCALER	ATIM _FM_009	Wrong or missing clock source for counter module	CPU registers, RAM: DC fault model for data and addresses	No or wrong frequency/duty PWM signal generation	P	-	-	ATIM_SM_3: Loopback scheme for PWM outputs
		ATIM _FM_010	Wrong or missing clock source for counter module	CPU registers, RAM: DC fault model for data and addresses	No or wrong counting function No or wrong interrupt generation on timer expiration	P	-	-	ATIM_SM_2: 1oo2 for input capture/counting timers NVIC_SM_1: Expected and unexpected interrupt check by application software
	AUTORELOAD	ATIM _FM_011	Missing or wrong autoreload/repetition function	CPU registers, RAM: DC fault model for data and addresses	No or wrong frequency/duty PWM signal generation	P	-	-	ATIM_SM_3: Loopback scheme for PWM outputs
		ATIM _FM_012		CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously)	T	-	-	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
General purpose / advanced timers	AUTORELOAD	ATIM _FM_013	Missing or wrong autoreload function	CPU registers, RAM: DC fault model for data and addresses	No or wrong counting function No or wrong interrupt generation on timer expiration	P	-	-	ATIM_SM_2: 1oo2 for input capture/counting timers NVIC_SM_1: Expected and unexpected interrupt check by application software
		ATIM _FM_014		CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously)	T	-	-	-
	CHANNEL(S) INPUT	ATIM _FM_015	No or wrong input signal for channel; wrong channel prescaler function	CPU registers, RAM: DC fault model for data and addresses	No or wrong input capture function	P	-	-	ATIM_SM_2: 1oo2 for input capture/counting timers
		ATIM _FM_016		CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously)	T	-	-	-
	CHANNEL(S) OUTPUT	ATIM _FM_017	No output propagation/generation	CPU registers, RAM: DC fault model for data and addresses	No PWM signal generation	P	-	-	ATIM_SM_3: Loopback scheme for PWM outputs
		ATIM _FM_018		CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously)	T	-	-	-
	CHANNEL(S) COMPARE REG	ATIM _FM_019	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No or wrong counting function No or wrong interrupt generation on timer expiration	P	-	-	ATIM_SM_2: 1oo2 for input capture/counting timers NVIC_SM_1: Expected and unexpected interrupt check by application software
		ATIM _FM_020			No or wrong frequency/duty PWM signal generation	P	-	-	ATIM_SM_3: Loopback scheme for PWM outputs
		ATIM _FM_021		CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously)	T	-	-	-
	CHANNEL(S) POLARITY	ATIM _FM_022	Wrong signal polarity selection	CPU registers, RAM: DC fault model for data and addresses	Wrong duty for PWM signal generation	P	-	-	ATIM_SM_3: Loopback scheme for PWM outputs
Basic timers	APB ITF	GTIM _FM_001	Data/address wrong values during TIM6/7 accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to TIM6/7 registers by application software	P	-	-	GTIM_SM_0: Periodical read-back of configuration registers
		GTIM _FM_002				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode – Detailed	Potential failure mode – IEC 61508	Potential effect(s) on the MCU	FM	Control – Prevention	Control – Detection	Recommended action(s)
Basic timers	CONFIG REGISTERS	GTIM _FM_003	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No or wrong counting function No or wrong interrupt generation on timer expiration	P	-	-	GTIM_SM_0: Periodical read-back of configuration registers NVIC_SM_1: Expected and unexpected interrupt check by application software
		GTIM _FM_004	-	CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	PRESCALER	GTIM _FM_005	Wrong or missing clock source for counter module	CPU registers, RAM: DC fault model for data and addresses		P	-	-	GTIM_SM_1: 1oo2 for counting timers NVIC_SM_1: Expected and unexpected interrupt check by application software
	AUTORELOAD	GTIM _FM_006	Missing or wrong autoreload/repetition function	CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously)	T	-	-	-
		GTIM _FM_007	Missing or wrong autoreload function	CPU registers, RAM: DC fault model for data and addresses	No or wrong counting function No or wrong interrupt generation on timer expiration	P	-	-	GTIM_SM_1: 1oo2 for counting timers NVIC_SM_1: Expected and unexpected interrupt check by application software
		GTIM _FM_008		CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously)	T	-	-	-
ADC	APB ITF	ADC _FM_001	Data/address wrong values during ADC accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to ADC peripheral by application software and/or DMA	P	-	-	ADC_SM_3: Periodical software test for ADC
		ADC _FM_002				T	-	-	ADC_SM_3: Periodical software test for ADC ADC_SM_1: Multiple acquisition by application software
		ADC _FM_003		Bus: No or continuous arbitration	No access to ADC peripheral by application software or DMA APB bus lock	P	-	-	ADC_SM_3: Periodical software test for ADC
	CONFIG REGISTERS	ADC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong configuration of ADC peripheral leading to wrong or missing ADC readings	P	-	-	ADC_SM_0: Periodical read-back of configuration registers
		ADC _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
ADC	TRIGGERS	ADC _FM_006	Wrong trigger source selection	CPU registers, RAM: DC fault model for data and addresses	No or wrong ADC conversion due to wrong acquisition/trigger timing	P	-	-	ADC_SM_3: Periodical software test for ADC
		ADC _FM_007		CPU registers, RAM: change of information caused by soft-errors		T	-	-	ADC_SM_1: Multiple acquisition by application software
	INPUT SEL	ADC _FM_008	Wrong ADC input source selection	CPU registers, RAM: DC fault model for data and addresses	No ADC correct signal acquisition Missing calibration operation	P	-	-	ADC_SM_3: Periodical software test for ADC ADC_SM_2: Range check by application software
		ADC _FM_009		CPU registers, RAM: change of information caused by soft-errors		T	-	-	ADC_SM_1: Multiple acquisition by application software ADC_SM_2: Range check by application software
	SAR	ADC _FM_010	Failure of SAR module (including analog section) or failure in reference voltages acquisition	CPU registers, RAM: DC fault model for data and addresses	Wrong analog value conversion	P	-	-	ADC_SM_3: Periodical software test for ADC ADC_SM_2: Range check by application software
		ADC _FM_011		CPU registers, RAM: change of information caused by soft-errors		T	-	-	ADC_SM_1: Multiple acquisition by application software ADC_SM_2: Range check by application software
	START/STOP	ADC _FM_012	Failure of start/stop conversion module	CPU registers, RAM: DC fault model for data and addresses	Missing ADC conversion	P	-	-	ADC_SM_2: Range check by application software NVIC_SM_1: Expected and unexpected interrupt check by application software
		ADC _FM_013			Missing interrupt generation	T	-	-	
	ANALOG WATCHDOG	ADC _FM_014	Failure of analog watchdog comparison function	CPU registers, RAM: DC fault model for data and addresses	Missing detection of out-of-range condition for input signal	P	-	-	ADC_SM_2: Range check by application software
		ADC _FM_015		CPU registers, RAM: change of information caused by soft-errors	Unexpected generation of out-of-range error	T	-	-	
	TEMP SENSOR	ADC _FM_016	Failure of internal temperature sensor	Analog I/O: DC fault model	Wrong internal temperature reading	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode – Detailed	Potential failure mode – IEC 61508	Potential effect(s) on the MCU	FM	Control – Prevention	Control – Detection	Recommended action(s)
DAC	APB ITF	DAC _FM_001	Data/address wrong values during DAC accesses by application software or DMA by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to DAC peripheral by application software and/or DMA	P	-	-	DAC_SM_0: Periodical read-back of configuration registers
		DAC _FM_002				T	-	-	DAC_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		DAC _FM_003		Bus: No or continuous arbitration	No access to DAC peripheral by application software or DMA APB bus lock	P	-	-	
	CONFIG REGISTERS	DAC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong configuration of DAC peripheral leading to wrong or missing DAC signal generation	P	-	-	DAC_SM_0: Periodical read-back of configuration registers
		DAC _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	CONTROL LOGIC	DAC _FM_006	Wrong DORx computation	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing DAC signal generation	P	-	-	DAC_SM_1: DAC output loopback on ADC channel
		DAC _FM_007		CPU registers, RAM: change of information caused by soft-errors	Mainly “no effect” (overwritten continuously refreshed	T	-	-	
		DAC _FM_008	Wrong or missing DMA request generation	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing DAC signal generation	P	-	-	
		DAC _FM_009	Wrong conversion	CPU registers, RAM: change of information caused by soft-errors	Mainly no effect (overwritten continuously refreshed	T	-	-	
	ANALOG SECTION	DAC _FM_010	Wrong analog signal generation	Analog I/O: DC fault model	Wrong or missing DAC signal generation	P	-	-	
IWDG	APB ITF	IWDG _FM_001	Data/address wrong values during IWDG accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to IWDG status/configuration registers by application software	P	-	-	WDG_SM_0: Periodical read-back of configuration registers
		IWDG _FM_002				T	-	-	
		IWDG _FM_003		Bus: No or continuous arbitration	No access to IWDG status/configuration registers by application software APB bus lock	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
IWDG	CONFIG REGISTERS	IWDG _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong IWDG configuration leading to watchdog disable	P	-	-	WDG_SM_0: Periodical read-back of configuration registers CPU_SM_5: External watchdog
		IWDG _FM_005	Wrong configuration register(s) value	CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		IWDG _FM_006	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong IWDG configuration leading to watchdog unexpected intervention	P	-	-	CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level
		IWDG _FM_007	Wrong configuration register(s) value	CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	PRESCALER	IWDG _FM_008	Wrong or missing counter/compare function management	CPU registers, RAM: DC fault model for data and addresses	Watchdog inability to generate reset signal under the related condition (missing trigger by application software)	P	-	-	CPU_SM_5: External watchdog
		IWDG _FM_009		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		IWDG _FM_010		CPU registers, RAM: DC fault model for data and addresses	Unexpected reset signal generation	P	-	-	CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level
		IWDG _FM_011		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
WWDG	APB ITF	WWDG _FM_001	Data/address wrong values during WWDG accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to WWDG status/configuration registers by application software	P	-	-	WDG_SM_0: Periodical read-back of configuration registers
		WWDG _FM_002				T	-	-	
		WWDG _FM_003	Data/address wrong values during WWDG accesses by application software by APB bus	Bus: No or continuous arbitration	No access to WWDG status/configuration registers by application software APB bus lock	P	-	-	
	CONFIG REGISTERS	WWDG _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong WWDG configuration leading to watchdog disable	P	-	-	WDG_SM_0: Periodical read-back of configuration registers CPU_SM_5: External watchdog
		WWDG _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	CONFIG REGISTERS	WWDG _FM_006	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong WWDG configuration leading to watchdog unexpected intervention	P	-	-	CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level
		WWDG _FM_007		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	PRESCALER	WWDG _FM_008	Wrong or missing counter/compare function management	CPU registers, RAM: DC fault model for data and addresses	Watchdog inability to generate reset signal under the related condition (missing trigger by application software)	P	-	-	CPU_SM_5: External watchdog
		WWDG _FM_009		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		WWDG _FM_010		CPU registers, RAM: DC fault model for data and addresses	Unexpected reset signal generation	P	-	-	CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level
		WWDG _FM_011		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
RTC	APB ITF	RTC _FM_001	Data/address wrong values during RTC accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to RTC status/configuration registers by application software	P	-	-	RTC_SM_0: Periodical read-back of configuration registers
		RTC _FM_002				T	-	-	
		RTC _FM_003		Bus: No or continuous arbitration	No or wrong access to RTC status/configuration registers by application software APB bus lock	P	-	-	
	CONFIG REGISTERS	RTC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong RTC configuration leading to wrong date/time readings	P	-	-	RTC_SM_0: Periodical read-back of configuration registers
		RTC _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		RTC _FM_006		CPU registers, RAM: DC fault model for data and addresses	Wrong RTC configuration leading to missing RTC-based interrupt generation	P	-	-	
		RTC _FM_007		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	BACKUP REGISTERS	RTC _FM_008	Wrong backup register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong values for data stored in backup registers	P	-	-	RTC_SM_2: Data redundancy methods on backup registers
		RTC _FM_009		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
RTC	COUNT SECTION	RTC _FM_010	Wrong clock source selection	CPU registers, RAM: DC fault model for data and addresses	Date and/or time wrongly updated	P	-	-	RTC_SM_1: Application check of running RTC
		RTC _FM_011	Wrong time counting function	CPU registers, RAM: DC fault model for data and addresses		P	-	-	
		RTC _FM_012		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		RTC _FM_013		CPU registers, RAM: DC fault model for data and addresses	Missing RTC-based interrupt generation	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
		RTC _FM_014		CPU registers, RAM: DC fault model for data and addresses	Continuous or unintended RTC-based interrupt generation	P	-	-	RTC_SM_1: Application check of running RTC NVIC_SM_1: Expected and unexpected interrupt check by application software
	TIMESTAMP	RTC _FM_015	Wrong timestamp acquisition function	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing acquisition of timestamp event	P	-	-	RTC_SM_3: Application-level measures to detect failures in timestamps/event capture
		RTC _FM_016		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	32 kHz XTAL	RTC _FM_017	Wrong, unstable or missing oscillation frequency	Analog I/O: DC fault model	Date and/or time wrongly updated	T	-	-	RTC_SM_1: Application check of running RTC

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
EXTI	APB ITF	EXTI _FM_001	Data/address wrong values during EXT-INT accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to EXT-INT peripheral by application software	P	-	-	EXTI_SM_0: Periodical read-back of configuration registers
EXTI	APB ITF	EXTI _FM_002	Data/address wrong values during EXT-INT accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to EXT-INT peripheral by application software	T	-	-	EXTI_SM_0: Periodical read-back of configuration registers
EXTI	APB ITF	EXTI _FM_003	Timeout or no arbitration in EXT-INT APB interface	Bus: No or continuous arbitration	No access to EXT-INT peripheral by application software APB bus lock	P	-	-	EXTI_SM_0: Periodical read-back of configuration registers
EXTI	REGISTERS	EXTI _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Missing external/internal interrupt generation Missing software interrupt generation	P	-	-	EXTI_SM_0: Periodical read-back of configuration registers
EXTI	REGISTERS	EXTI _FM_005	Wrong configuration register(s) value	CPU registers, RAM: change of information caused by soft-errors	Missing external/internal interrupt generation	T	-	-	EXTI_SM_0: Periodical read-back of configuration registers
EXTI	-	EXTI _FM_006	Failure in masking and/or edge detection circuits	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Missing interrupt generation for internal hardware source	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
EXTI	-	EXTI _FM_007	Failure in masking and/or edge detection circuits	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Missing interrupt generation for internal hardware source	T	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
EXTI	-	EXTI _FM_008	Continuous or unintended interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software CPU_SM_1: Control flow monitoring in application software
SYSCFG	APB ITF	SYSCFG _FM_001	Data/address wrong values during SYSCFG accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to SYSCFG peripheral by application software	P	-	-	SYSCFG_SM_0: Periodical read-back of configuration registers
SYSCFG	APB ITF	SYSCFG _FM_002	Data/address wrong values during SYSCFG accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to SYSCFG peripheral by application software	T	-	-	SYSCFG_SM_0: Periodical read-back of configuration registers
SYSCFG	APB ITF	SYSCFG _FM_003	Timeout or no arbitration in SYSCFG APB interface	Bus: No or continuous arbitration	No access to SYSCFG peripheral by application software	P	-	-	SYSCFG_SM_0: Periodical read-back of configuration registers

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
SYSCFG	REGISTERS	SYSCFG_FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong DMA channels settings Wrong interrupt configuration for GPIO Wrong management of pending interrupts Wrong Flash memory /RAM mapping Wrong Comparator settings	P	-	-	SYSCFG_SM_0: Periodical read-back of configuration registers
		SYSCFG_FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	SYSCFG_SM_0: Periodical read-back of configuration registers NVIC_SM_1: Expected and unexpected interrupt check by application software
CRC	APB ITF	CRC_FM_001	Data/address wrong values during CRC accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to CRC peripheral by application software	P	-	-	CRC_SM_0: CRC self-coverage
		CRC_FM_002				T	-	-	
		CRC_FM_003	Timeout or no arbitration in CRC APB interface	Bus: No or continuous arbitration	No access to CRC peripheral by application software APB bus lock	P	-	-	CRC_SM_0: CRC self-coverage BUS_SM_0: Periodical software test for interconnections
	REGISTERS	CRC_FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong CRC computation due to wrong polynomial seed Wrong CRC computed value	P	-	-	CRC_SM_0: CRC self-coverage
		CRC_FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	-	CRC_FM_006	Failures of CRC computation engine	CPU registers, RAM: DC fault model for data and addresses	Wrong CRC computed value	P	-	-	
		CRC_FM_007		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode - Detailed	Potential failure mode - IEC 61508	Potential effect(s) on the MCU	FM	Control - Prevention	Control - Detection	Recommended action(s)
RCC	APB ITF	RCC _FM_001	Data/address wrong values during RCC accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to RCC peripheral by application software	P	-	-	CLK_SM_0: Periodical read-back of configuration registers
		RCC _FM_002				T	-	-	
		RCC _FM_003	Timeout or no arbitration in RCC APB interface	Bus: No or continuous arbitration	No access to RCC peripheral by application software APB bus lock	P	-	-	
	REGISTERS	RCC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	PLL/Clock sources wrong setting	P	-	-	CPU_SM_5: External watchdog CLK_SM_0: Periodical read-back of configuration registers
		RCC _FM_005		CPU registers, RAM: change of information caused by soft-errors	Clock stabilization-related interrupt wrong setting	T	-	-	
		RCC _FM_006		CPU registers, RAM: DC fault model for data and addresses	CSS clock security system disable	P	-	-	CLK_SM_0: Periodical read-back of configuration registers
		RCC _FM_007		CPU registers, RAM: change of information caused by soft-errors	Power and watchdog -related reset disable	T	-	-	
		RCC _FM_008		CPU registers, RAM: DC fault model for data and addresses	Missing or unintended peripheral reset	P	-	-	
		RCC _FM_009		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
		RCC _FM_010		CPU registers, RAM: DC fault model for data and addresses	Unintended enable or disable clock source for one or more peripherals	P	-	-	
		RCC _FM_011		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
RCC	CONTROL LOGIC	RCC _FM_012	Failure of reset control and distribution logic	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Missing or unintended peripheral reset	P	-	-	CoU_4: End User must implement the required combination of safety mechanism/CoUs for each STM32 peripherals used in safety function(s) implementation
		RCC _FM_013				T	-	-	
		RCC _FM_014	Failure of clock distribution logic		Unintended enable or disable clock source for one or more peripherals	P	-	-	
		RCC _FM_015				T	-	-	
Clocks	HSE oscillator	CLOCKS _FM_001	No clock signal	Clock: Incorrect frequency	No program execution	P	-	CLK_SM_1: CSS Clock Security System	CPU_SM_5: External watchdog
		CLOCKS _FM_002	Clock frequency lower than the specified one		Application software executed at lower speed No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CLOCKS _FM_003	Clock frequency higher than the specified one		Application software executed at higher speed so with wrong timings No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software CoU_4: End User must implement the required combination of safety mechanism/CoUs for each STM32 peripherals used in safety function(s) implementation
		CLOCKS _FM_004	Jitter on clock frequency	Clock: Period jitter	Application software executed at wrong and non constant speed Sporadic communication errors affecting communication peripherals	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
Clocks	RC 8 MHz oscillator	CLOCKS _FM_005	No clock signal	Clock: Incorrect frequency	No program execution	P	-	CLK_SM_1: CSS Clock Security System	CPU_SM_5: External watchdog
		CLOCKS _FM_006	Clock frequency lower than the specified one	Clock: Incorrect frequency	Application software executed at lower speed No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CLOCKS _FM_007	Clock frequency higher than the specified one	Clock: Incorrect frequency	Application software executed at higher speed so with wrong timings No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software
		CLOCKS _FM_008	Jitter on clock frequency	Clock: Period jitter	Application software executed at wrong and non constant speed Sporadic communication errors affecting communication peripherals	P	-	-	
	RC 14 MHz oscillator	CLOCKS _FM_009	No clock signal	Clock: Incorrect frequency	No program execution	P	-	CLK_SM_1: CSS Clock Security System	CPU_SM_5: External watchdog
		CLOCKS _FM_010	Clock frequency lower than the specified one		Application software executed at lower speed No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CLOCKS _FM_011			Application software executed at higher speed so with wrong timings No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software CoU_4: End User must implement the required combination of safety mechanism/CoUs for each STM32 peripherals used in safety function(s) implementation
		CLOCKS _FM_012	Jitter on clock frequency	Clock: Period jitter	Application software executed at wrong and non constant speed Sporadic communication errors affecting communication peripherals	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
Clocks	PLL	CLOCKS _FM_013	No clock signal	Clock: Incorrect frequency	No program execution	P	-	CLK_SM_1: CSS Clock Security System	CPU_SM_5: External watchdog
		CLOCKS _FM_014	Clock frequency lower than the specified one		Application software executed at lower speed No communication over communication peripherals	P	-	-	CLK_SM_3: Internal clock cross-measure CLK_SM_2: Independent Watchdog
		CLOCKS _FM_015	Clock frequency higher than the specified one		Application software executed at higher speed so with wrong timings No communication over communication peripherals	P	-	-	CLK_SM_3: Internal clock cross-measure CoU_4: End User must implement the required combination of safety mechanism/CoUs for each STM32 peripherals used in safety function(s) implementation
		CLOCKS _FM_016	Jitter on clock frequency	Clock: Period jitter	Application software executed at wrong and non constant speed Sporadic communication errors affecting communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software CoU_4: End User must implement the required combination of safety mechanism/CoUs for each STM32 peripherals used in safety function(s) implementation
	PRESCALERS	CLOCKS _FM_017	No clock source selected	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No program execution	P	-	-	CPU_SM_5: External watchdog
		CLOCKS _FM_018	Wrong clock source selected or wrong prescaling operation		Application software executed at lower speed No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CLOCKS _FM_019			Application software executed at higher speed so with wrong timings No communication over communication peripherals	P	-	-	CPU_SM_1: Control flow monitoring in application software
		CLOCKS _FM_020			Sporadic perturbation of internal clock speed	T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Power	REG33TO18	POWER _FM_001	1.8 Voltage line value continuously above permitted specified range	Power supply: DC fault model	No program execution Wrong program execution External communication failures	P	-	-	VSUP_SM_2: Independent Watchdog VSUP_SM_3: Internal temperature sensor check
		POWER _FM_002	1.8 Voltage line value continuously below permitted specified range			P	-	-	VSUP_SM_1: Supply voltage internal monitoring (PVD) VSUP_SM_2: Independent Watchdog
		POWER _FM_003	1.8 Voltage line sporadic excursions outside of permitted specified range	Power supply: Drift and oscillation		P	-	-	
		POWER _FM_004	1.8 Voltage absent due to regulator unintended switch-off		No program execution	P	-	-	VSUP_SM_2: Independent Watchdog
	SWITCH	POWER _FM_005	Unintended interruption of Backup circuitry section supply	Analog I/O: DC fault model	RTC not working Backup registers contents compromised	P	-	-	RTC_SM_2: Data redundancy methods on backup registers RTC_SM_1: Application check of running RTC
		POWER _FM_006	Missing switch to VBAT supply for Backup circuitry section		RTC not working during no-power phase	P	-	-	CoU_3: Low power mode state must not be used in safety function(s) implementation
		POWER _FM_007			Backup registers contents compromised	P	-	-	RTC_SM_2: Data redundancy methods on backup registers CoU_3: Low power mode state must not be used in safety function(s) implementation
	CONTROL	POWER _FM_008	Missed switch to stop/low power mode	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No low power/stop state achievable	P	-	-	CPU_SM_1: Control flow monitoring in application software
		POWER _FM_009				T	-	-	
		POWER _FM_010	Unintended switch to Stop / Low power mode		No program execution	P	-	-	CPU_SM_5: External watchdog
		POWER _FM_011				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
Power	REGISTERS	POWER _FM_012	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Wrong power options settings	P	-	-	VSUP_SM_0: Periodical read-back of configuration registers CPU_SM_5: External watchdog
		POWER _FM_013		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	POR	POWER _FM_014	Missing release of reset at startup	Analog I/O: DC fault model	No program execution	P	-	-	CPU_SM_5: External watchdog
		POWER _FM_015	Unintended release of reset during operation			P	-	-	
		POWER _FM_016	Early release of reset during startup		Wrong CPU behaviour Wrong application software execution	P	-	-	CPU_SM_5: External watchdog CPU_SM_1: Control flow monitoring in application software
	PDR	POWER _FM_017	Missing release of reset at power-down		No program execution	P	-	-	CPU_SM_5: External watchdog
		POWER _FM_018	Unintended release of reset during operation		No program execution	P	-	-	
	PVD	POWER _FM_019	Failure in masking and/or edge detection circuits		No protection against voltage wrong values	P	-	-	
		POWER _FM_020	Unintended reset generation during operation		No program execution	P	-	-	CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
Debug MCU	APB ITF	DEBUG _FM_001	Data/address wrong values during DEBUGMCU accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to DEBUGMCU peripheral by application software	P	-	-	COMP_SM_0: Periodical read-back of configuration registers
		DEBUG _FM_002			No or wrong access to DEBUGMCU peripheral by application software APB bus lock	T	-	-	DBG_SM_0: Periodical read-back of configuration registers BUS_SM_0: Periodical software test for interconnections
		DEBUG _FM_003	Timeout or no arbitration in DEBUGMCU APB interface	Bus: No or continuous arbitration	No access to DEBUGMCU peripheral by application software APB bus lock	P	-	-	BUS_SM_0: Periodical software test for interconnections
	REGISTERS	DEBUG _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Unintended debug activation leading to no program execution (CPU halted)	P	-	-	DBG_SM_0: Periodical read-back of configuration registers
		DEBUG _FM_005		CPU registers, RAM: change of information caused by soft-errors	Internal timer stop counting Window watchdog disable	T	-	-	
	-	DEBUG _FM_006	Unintended debug activation generating CPU reset/Halt/Sleep	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No program execution	P	-	-	CPU_SM_5: External watchdog
		DEBUG _FM_007				T	-	-	
		DEBUG _FM_008	Wrong generation of internal debug signals controlling timers		Internal timer stop counting	P	-	-	ATIM_SM_1: 1oo2 for counting timers
		DEBUG _FM_009	Wrong generation of internal debug signals controlling window watchdog counter		Window watchdog disable	P	-	-	CPU_SM_5: External watchdog
		DEBUG _FM_010	Failures of debug internal logic		Mainly no effect (debug not used)	P	-	-	CPU_SM_5: External watchdog CoU_2: STM32F4 debug features must not be used in safety function(s) implementation
		DEBUG _FM_011				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
beCAN	APB ITF	CAN _FM_001	Data/address wrong values during CAN accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to CAN peripheral by application software and/or DMA	P	-	-	CAN_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		CAN _FM_002	Data/address wrong values during CAN accesses by application software and/or DMA by APB bus			T	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing DMA_SM_4: DMA transaction awareness checks
		CAN _FM_003		Bus: No or continuous arbitration	No access to CAN peripheral by application software or DMA APB bus lock	P	-	-	CAN_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	CAN _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No CAN communication Incoming message lost due to wrong filtering RX buffer overrun due to missing interrupt generation	P	-	-	CAN_SM_0: Periodical read-back of configuration registers
		CAN _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	INTERRUPT CTRL	CAN _FM_006	Missing interrupt generation on receive/transmit data	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No CAN data processing by application software	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing NVIC_SM_1: Expected and unexpected interrupt check by application software
		CAN _FM_007			Missing transmission of CAN data RX mailbox overrun	T	-	-	
		CAN _FM_008	Continuous or unintended CAN interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software
	OUTPUTS	CAN _FM_009	TX and/or RX signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing CAN data received Wrong or missing CAN data transmitted	P	-	CAN_SM_1: Protocol error signals	CAN_SM_2: Information redundancy techniques on messages, including End to End safing

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
beCAN	CONTROL	CAN _FM_010	Wrong data processing in receive/transmit processing control units	CPU: Coding and execution including flag registers, Wrong coding or wrong execution	Wrong or missing CAN data received	P	-	CAN_SM_1: Protocol error signals	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
		CAN _FM_011			Wrong or missing CAN data transmitted	T	-		
		CAN _FM_012	Missed CAN wakeup on transmission/receive request		Missing CAN data received Missing CAN data transmitted	P	-	-	
	MAC	CAN _FM_013	Wrong message management into Rx/Tx mailboxes		Wrong or missing CAN messages received	P	-	-	
		CAN _FM_014			Wrong or missing CAN messages transmitted	T	-	-	
	INTERNAL RAM	CAN _FM_015	Wrong values in one or several bits in internal RAM buffer	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing CAN messages received	P	-	-	
		CAN _FM_016		CPU registers, RAM: change of information caused by soft-errors	Wrong or missing CAN messages transmitted	T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
beCAN	-	CAN _FM_017	CAN communication affected by transmission errors	Data communication: transmission errors	Wrong CAN data received Wrong CAN data transmitted	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
beCAN	-	CAN _FM_018	CAN communication affected by transmission repetition	Data communication: transmission repetitions	Wrong CAN data received Wrong CAN data transmitted	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
beCAN	-	CAN _FM_019	CAN communication affected by transmission deletion	Data communication: transmission deletion	Wrong CAN data received Wrong CAN data transmitted	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
beCAN	-	CAN _FM_020	CAN communication affected by transmission resequencing	Data communication: transmission resequencing	Message sent or received in wrong order	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
beCAN	-	CAN _FM_021	CAN communication affected by transmission corruption	Data communication: transmission corruption	Wrong CAN data received Wrong CAN data transmitted	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
beCAN	-	CAN _FM_022	CAN communication affected by transmission delay	Data communication: transmission delay	Wrong or missing CAN data received Wrong or missing CAN data transmitted	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
beCAN	-	CAN _FM_023	CAN communication affected by transmission masquerading	Data communication: transmission masquerading	Unintended message reception	P	-	-	CAN_SM_2: Information redundancy techniques on messages, including End to End safing
Level shifters	V33/V18	LVLSHT _FM_001	Wrong level translation from V33 to V18 domain (and vice-versa)	Analog I/O: DC fault model	Wrong power configuration	P	-	-	CPU_SM_5: External watchdog
Level shifters	V33/V18	LVLSHT _FM_002	Wrong level translation from V33 to V18 domain (and vice-versa)	Analog I/O: DC fault model	Wrong clock configuration	P	-	-	CPU_SM_5: External watchdog
Level shifters	V33/V18	LVLSHT _FM_003	Wrong level translation from V33 to V18 domain (and vice-versa)	Analog I/O: DC fault model	Missing or wrong access to RTC functions (backup registers, calendar)	P	-	-	RTC_SM_1: Application check of running RTC RTC_SM_2: Data redundancy methods on backup registers

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode - Detailed	Potential failure mode - IEC 61508	Potential effect(s) on the MCU	FM	Control - Prevention	Control - Detection	Recommended action(s)
Level shifters	V18/VDDA	LVLSHT _FM_004	Wrong level translation from V18 to VDDA domains (and vice-versa)	Analog I/O: DC fault model	Wrong ADC reading	P	-	-	ADC_SM_2: Range check by application software ADC_SM_3: Periodical software test for ADC
		LVLSHT _FM_005			Wrong DAC signal generation	P	-	-	DAC_SM_1: DAC output loopback on ADC channel
Temperature sensor	-	TMP _FM_001	Wrong junction temperature sensing	Analog I/O: DC fault model	Junction temperature measure not accurate	P	-	-	TMP_SM_0: Plausibility check on temperature sensor value
CRS	APB ITF	CRS _FM_001	Data/address wrong values during CRS accesses by application software by APB bus	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to CRS peripheral by application software	P	-	-	CRS_SM_0: Periodical read-back of configuration registers
		CRS _FM_002		CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No or wrong access to CRS peripheral by application software	T	-	-	
		CRS _FM_003		Bus: No or continuous arbitration	No access to CRS peripheral by application software APB bus lock	P	-	-	
	REGISTERS	CRS _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	CRS misconfiguration CRS detection disable CRS thresholds wrong programming	P	-	-	
		CRS _FM_005	Wrong configuration register(s) value	CPU registers, RAM: change of information caused by soft-errors	-	T	-	-	
	INTERRUPT CTRL	CRS _FM_006	Missing interrupt generation on CRS detected clock timing violation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	No clock error detection reporting	P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CRS _FM_007				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
CRS	INTERRUPT CTRL	CRS _FM_008	Unintended interrupt generation on CRS detected clock timing violation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Unintended clock error detection reporting	P	-	CRS_SM_1: CRC check for clock frequency errors	-
		CRS _FM_009				T	-		-
		CRS _FM_010	Continuous or unintended CRS interrupt request generation	Interrupt: no or continuous interrupt		P	-		-
	CONTROL	CRS _FM_011	Wrong clock comparison leading to wrong error reporting/generation	CPU: Coding and execution including flag registers. Wrong coding or wrong execution	Unintended clock error detection reporting	P	-		-
		CRS _FM_012				T	-		-
		CRS _FM_013	Wrong clock comparison leading to missing error reporting/generation		No clock error detection reporting	P	-	-	CPU_SM_1: Control flow monitoring in application software CPU_SM_5: External watchdog
		CRS _FM_014				T	-	-	
SD/SDIO/MMC	APB_ITF	SDMMC _FM_001	Data/address wrong values during SDMMC accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to SDMMC peripheral by application software and/or DMA	P	-	-	SDMMC_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		SDMMC _FM_002	Data/address wrong values during SDMMC accesses by application software and/or DMA by APB bus			T	-	-	SDMMC_SM_2: Information redundancy techniques on messages DMA_SM_4: DMA transaction awareness checks
		SDMMC _FM_003		Bus: No or continuous arbitration	No access to SDMMC peripheral by application software or DMA APB bus lock	P	-	-	SDMMC_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	SDMMC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No SDMMC communication	P	-	-	SDMMC_SM_0: Periodical read-back of configuration registers
		SDMMC _FM_005				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
SD/SDIO/MMC	INTERRUPT CTRL	SDMMC_FM_006	Missing interrupt generation on receive/transmit data	CPU coding and execution including flag registers: Wrong coding or wrong execution	No SDMMC data processing by application software	P	-	-	SDMMC_SM_2: Information redundancy techniques on messages NVIC_SM_1: Expected and unexpected interrupt check by application software
SD/SDIO/MMC	INTERRUPT CTRL	SDMMC_FM_007	Missing interrupt generation on receive/transmit data	CPU coding and execution including flag registers: Wrong coding or wrong execution	No SDMMC data processing by application software	T	-	-	SDMMC_SM_2: Information redundancy techniques on messages NVIC_SM_1: Expected and unexpected interrupt check by application software
SD/SDIO/MMC	INTERRUPT CTRL	SDMMC_FM_008	Continuous or unintended SDMMC interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software
SD/SDIO/MMC	OUTPUTS	SDMMC_FM_009	TX and/or RX signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing SDMMC data received Wrong or missing SDMMC data transmitted	P	-	SDMMC_SM_1: Protocol error signals and information redundancy including hardware CRC	-
SD/SDIO/MMC	CONTROL	SDMMC_FM_010	Wrong data processing in receive/transmit processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing SDMMC data received	P	-	SDMMC_SM_1: Protocol error signals and information redundancy including hardware CRC	-
SD/SDIO/MMC	CONTROL	SDMMC_FM_011	Wrong data processing in receive/transmit processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing SDMMC data transmitted	T	-	SDMMC_SM_1: Protocol error signals and information redundancy including hardware CRC	-
SD/SDIO/MMC	CONTROL	SDMMC_FM_012	Missed SDMMC wakeup on transmission/receive request	CPU coding and execution including flag registers: Wrong coding or wrong execution	Missing SDMMC data received Missing SDMMC data transmitted	P	-	SDMMC_SM_1: Protocol error signals and information redundancy including hardware CRC	SDMMC_SM_2: Information redundancy techniques on messages
SD/SDIO/MMC	-	SDMMC_FM_017	SDMMC communication affected by transmission errors	Data communication: transmission errors	Wrong SDMMC data received Wrong SDMMC data transmitted	P	-	SDMMC_SM_1: Protocol error signals and information redundancy including hardware CRC	SDMMC_SM_2: Information redundancy techniques on messages
SD/SDIO/MMC	-	SDMMC_FM_018	SDMMC communication affected by transmission repetition	Data communication: transmission repetitions	Wrong SDMMC data received Wrong SDMMC data transmitted	P	-	-	SDMMC_SM_2: Information redundancy techniques on messages
SD/SDIO/MMC	-	SDMMC_FM_019	SDMMC communication affected by transmission deletion	Data communication: transmission deletion	Missing SDMMC data received Missing SDMMC data transmitted	P	-	SDMMC_SM_1: Protocol error signals and information redundancy including hardware CRC	SDMMC_SM_2: Information redundancy techniques on messages

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
SD/SDIO/MMC	-	SDMMC _FM_020	SDMMC communication affected by transmission resequencing	Data communication: transmission resequencing	Message sent or received in wrong order	P	-	-	SDMMC_SM_2: Information redundancy techniques on messages
		SDMMC _FM_021	SDMMC communication affected by transmission corruption	Data communication: transmission corruption	Wrong SDMMC data received Wrong SDMMC data transmitted	P	-	SDMMC_SM_1: Protocol error signals and information redundancy including hardware CRC	
		SDMMC _FM_022	SDMMC communication affected by transmission delay	Data communication: transmission delay	Wrong or missing SDMMC data received Wrong or missing SDMMC data transmitted	P	-		
		SDMMC _FM_023	SDMMC communication affected by transmission masquerading	Data communication: transmission masquerading	Unintended message reception	P	-	-	
RNG	APB ITF	RNG _FM_001	Data/address wrong values during RNG accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to RNG peripheral by application software and/or DMA	P	-	-	RNG_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		RNG _FM_002	Data/address wrong values during RNG accesses by application software and/or DMA by APB bus	Bus: No or continuous arbitration	No or wrong access to RNG peripheral by application software and/or DMA	T	-	-	RNG_SM_1: RNG module entropy on-line tests
		RNG _FM_003			No access to RNG peripheral by application software or DMA APB bus lock	P	-	-	RNG_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
RNG	REGISTERS	RNG _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	RNG misconfiguration	P	-	-	RNG_SM_0: Periodical read-back of configuration registers
		RNG _FM_005	Wrong configuration register(s) value	CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	CONTROL	RNG _FM_006	Missing interrupt generation on receive/transmit data	CPU coding and execution including flag registers: Wrong coding or wrong execution	No RNG true random data generation (i.e. single repetition)	P	-	-	RNG_SM_1: RNG module entropy on-line tests
		RNG _FM_007				T	-	-	
CRYP	APB ITF	CRYP _FM_001	Data/address wrong values during CRYP accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to CRYP peripheral by application software and/or DMA	P	-	-	CRYP_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		CRYP _FM_002	Data/address wrong values during CRYP accesses by application software and/or DMA by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to CRYP peripheral by application software and/or DMA	T	-	-	CRYP_SM_1: Encryption/decryption collateral detection DMA_SM_4: DMA transaction awareness checks
		CRYP _FM_003		Bus: No or continuous arbitration	No access to CRYP peripheral by application software or DMA APB bus lock	P	-	-	CRYP_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	CRYP _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	CRYP misconfiguration	P	-	-	CRYP_SM_0: Periodical read-back of configuration registers
		CRYP _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	INTERRUPT CTRL	CRYP _FM_006	Missing interrupt generation on receive/transmit/ processing data	CPU coding and execution including flag registers: Wrong coding or wrong execution	No CRYP data processing by application software and/or no CRYP operation report	P	-	-	CRYP_SM_2: Information redundancy techniques on messages to implement full End to End safing NVIC_SM_1: Expected and unexpected interrupt check by application software
		CRYP _FM_007				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode – Detailed	Potential failure mode – IEC 61508	Potential effect(s) on the MCU	FM	Control – Prevention	Control – Detection	Recommended action(s)
CRYP	INTERRUPT CTRL	CRYP _FM_008	Continuous or unintended CRYP interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software
	CONTROL	CRYP _FM_009	Wrong data processing in cypher/decypher processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong cypher operation	P	-	-	CRYP_SM_1: Encryption/decryption collateral detection
		CRYP _FM_010			Wrong decypher operation	T	-	-	CRYP_SM_1: Encryption/decryption collateral detection
		CRYP _FM_011			Message data corruption/alteration	P	-	-	CRYP_SM_2: Information redundancy techniques on messages to implement full End to End safing
		CRYP _FM_012				T	-	-	
HASH	APB ITF	HASH _FM_001	Data/address wrong values during HASH accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to HASH peripheral by application software and/or DMA	P	-	-	HASH_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		HASH _FM_002	Data/address wrong values during HASH accesses by application software and/or DMA by APB bus			T	-	-	HASH_SM_1: HASH processing collateral detection DMA_SM_4: DMA transaction awareness checks
		HASH _FM_003		Bus: No or continuous arbitration	No access to HASH peripheral by application software or DMA APB bus lock	P	-	-	HASH_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	HASH _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	HASH misconfiguration	P	-	-	HASH_SM_0: Periodical read-back of configuration registers
		HASH _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
HASH	INTERRUPT CTRL	HASH _FM_006	Missing interrupt generation on receive/transmit/processing data	CPU coding and execution including flag registers: Wrong coding or wrong execution	No HASH data processing by application software and/or no HASH operation report	P	-	-	HASH_SM_1: HASH processing collateral detection NVIC_SM_1: Expected and unexpected interrupt check by application software
		HASH _FM_007				T	-	-	
		HASH _FM_008	Continuous or unintended HASH interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software
	CONTROL	HASH _FM_009	Wrong data processing in cypher/decypher processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong HASH coding/encoding	P	-	-	HASH_SM_1: HASH processing collateral detection
		HASH _FM_010				T	-	-	
		HASH _FM_011			Message data corruption/alteration	P	-	-	
		HASH _FM_012				T	-	-	
DCMI	APB ITF	DCMI _FM_001	Data/address wrong values during DCMI accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to DCMI peripheral by application software and/or DMA	P	-	-	DCMI_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		DCMI _FM_002	Data/address wrong values during DCMI accesses by application software and/or DMA by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to DCMI peripheral by application software and/or DMA	T	-	-	
		DCMI _FM_003		Bus: No or continuous arbitration	No access to DCMI peripheral by application software/DMA or APB bus lock	P	-	-	
	REGISTERS	DCMI _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	DCMI misconfiguration	P	-	-	DCMI_SM_0: Periodical read-back of configuration registers
		DCMI _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
DCMI	CONTROL	DCMI _FM_006	No data video stream received or wrong format conversion	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing data stream received	P	-	DCMI_SM_1: DCMI video input data synchronization	-
		DCMI _FM_007	Data errors in received data stream		Wrong data in video stream received	T	-		-
LTDC	APB ITF	LTDC _FM_001	Data/address wrong values during LCD accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to LCD peripheral by application software and/or DMA	P	-	-	LCD_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		LTDC _FM_002	Data/address wrong values during LCD accesses by application software and/or DMA by APB bus			T	-	-	
		LTDC _FM_003		Bus: No or continuous arbitration	No access to LCD peripheral by application software/DMA APB bus lock	P	-	-	
	REGISTERS	LTDC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	LCD misconfiguration	P	-	-	LCD_SM_0: Periodical read-back of configuration registers
		LTDC _FM_005	Wrong configuration register(s) value	CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	CONTROL	LTDC _FM_006	Wrong LCD signal generation/coding	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong LCD image	P	-	-	LCD_SM_1: LCD acquisition by ADC channel
		LTDC _FM_007			Glitch in LCD image	T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
FMC	APB ITF	FMC _FM_001	Data/address wrong values during DCMI accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to DCMI peripheral by application software and/or DMA	P	-	-	-
		FMC FM_002	Data/address wrong values during DCMI accesses by application software and/or DMA by APB bus			T	-	-	-
		FMC _FM_003		Bus: No or continuous arbitration	No access to DCMI peripheral by application software/DMA APB bus lock	P	-	-	-
	REGISTERS	FMC _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	Misconfiguration leading to: – wrong application data – wrong application software execution	P	-	-	-
		FMC _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	-
	CONTROLLER	FMC _FM_006	Instruction read form wrong address and/or error in instruction code read	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong application software execution	P	-	FSMC_SM_3: ECC engine on NAND interface in FSMC module	FSMC_SM_0: Control flow monitoring in application software FSMC_SM_1: Information redundancy on external memory connected to FSMC
		FMC _FM_007				T	-		
		FMC _FM_008	Data read/write from/to wrong address and/or data value read/write error		Wrong application data	P	-		
		FMC _FM_009				T	-		
	ECC engine	FMC _FM_010	Wrong data decoding without error reporting		Wrong application software execution	P	-	-	
		FMC _FM_011			Wrong application data	T	-	-	
		FMC _FM_012	Unintended or continuous error detection interrupt generation		Unintended error reporting	P	-	-	CoU_1: The reset condition of Arm® Cortex® M4 CPU must be compatible as valid safe state at system level
		FMC _FM_013				T	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
FMC	OUTPUTS	FMC _FM_014	FMC connection signals wrong value, open or HiZ	Digital I/O: DC fault model	Misconfiguration leading to: – wrong application data – wrong application software execution	P	-	-	FSMC_SM_0: Control flow monitoring in application software FSMC_SM_1: Information redundancy on external memory connected to FSMC
DSI	APB ITF	DSI _FM_001	Data/address wrong values during LCD accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to LCD peripheral by application software and/or DMA	P	-	-	DSI_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		DSI _FM_002	Data/address wrong values during LCD accesses by application software and/or DMA by APB bus			P	-	-	
		DSI _FM_003		Bus: No or continuous arbitration	No access to LCD peripheral by application software/DMA APB bus lock	P	-	-	
	REGISTERS	DSI _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	DSI interface misconfiguration leading to no image	P	-	-	DSI_SM_0: Periodical read-back of configuration registers
		DSI _FM_005	Wrong configuration register(s) value	CPU registers, RAM: change of information caused by soft-errors	DSI interface misconfiguration leading to no image	T	-	-	
	CONTROL	DSI _FM_006	Wrong DSI data stream generation/coding	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing image on DSI connected device	P	-	DSI_SM_1: Protocol error signals and information redundancy including hardware checksums	-
		DSI _FM_007			Wrong image on DSI connected device	T	-		-
	OUTPUTS	DSI _FM_008	DSI signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing image on DSI connected device	P	-		-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
ETH	APB ITF	ETH _FM_001	Data/address wrong values during ETH accesses by application software and/or DMA by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to ETH peripheral by application software and/or DMA	P	-	-	ETH_SM_0: Periodical read-back of configuration registers
		ETH _FM_002				T	-	-	ETH_SM_2: Information redundancy techniques on messages, including End to End safing DMA_SM_4: DMA transaction awareness checks
		ETH _FM_003	Timeout or no arbitration in ETH APB interface	Bus: No or continuous arbitration	No access to ETH peripheral by application software or DMA APB bus lock	P	-	-	ETH_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	ETH _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No ETH communication	P	-	-	ETH_SM_0: Periodical read-back of configuration registers
		ETH _FM_005		CPU registers, RAM: change of information caused by soft-errors		T	-	-	
	BAUD GENERATOR	ETH _FM_006	Wrong or missing clock generation	CPU coding and execution including flag registers: Wrong coding or wrong execution		P	-	-	ETH_SM_2: Information redundancy techniques on messages, including End to End safing
	INTERRUPT CTRL	ETH _FM_007	Missing interrupt generation on receive/transmit data		No ETH data processing by application software Missing transmission of ETH data	P	-	-	ETH_SM_2: Information redundancy techniques on messages, including End to End safing NVIC_SM_1: Expected and unexpected interrupt check by application software
		ETH _FM_008		CPU coding and execution including flag registers: Wrong coding or wrong execution	No ETH data processing by application software Missing transmission of ETH data ETH receive buffer overrun	T	-	-	
		ETH _FM_009	Continuous or unintended ETH interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
ETH	OUTPUTS	ETH _FM_010	MDC and/or MIO signals wrong value, open or HiZ	Digital I/O: DC fault model	Missing or wrong ETH data communication	P	-	ETH_SM_1: Protocol error signals including hardware CRC	-
		ETH _FM_011	MII signals wrong value, open or HiZ			P	-		-
	CONTROL	ETH _FM_012	Wrong data processing in receive/transmit processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing ETH data received	P	-		ETH_SM_2: Information redundancy techniques on messages, including End to End safing
		ETH _FM_013				T	-		
	FIFOS	ETH _FM_014	Single or multiple bit errors in Rx/Tx FIFO cells	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing ETH data transmitted	P	-		-
		ETH _FM_015		CPU registers, RAM: change of information caused by soft-errors		T	-		-
	CRC CONTROLLER	ETH _FM_016	Wrong CRC computation for outgoing message	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong CRC field in outgoing message	P	-		-
		ETH _FM_017				P	-		-
		ETH _FM_018	Wrong CRC computation for incoming message		CRC mismatch in well-received message	P	-		-
		ETH _FM_019				P	-		-
		ETH _FM_020	CRC error reporting disable		CRC error protection disable	P	-	-	ETH_SM_2: Information redundancy techniques on messages, including End to End safing
		ETH _FM_021	Unintended CRC error reporting		CRC error propagated so system will switch in safe state	P	-	ETH_SM_1: Protocol error signals including hardware CRC	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
ETH	-	ETH _FM_022	ETH communication affected by transmission errors	Data communication: transmission errors	Wrong ETH data received Wrong ETH data transmitted	P	-	ETH_SM_1: Protocol error signals including hardware CRC	-
		ETH _FM_023	ETH communication affected by transmission repetitions	Data communication: transmission repetitions		P	-		ETH_SM_2: Information redundancy techniques on messages, including End to End safing
		ETH _FM_024	ETH communication affected by transmission deletion	Data communication: transmission deletion		P	-		-
		ETH _FM_025	ETH communication affected by transmission resequencing	Data communication: transmission resequencing	ETH data received in wrong sequence ETH data transmitted in wrong sequence	P	-		-
		ETH _FM_026	ETH communication affected by transmission corruption	Data communication: transmission corruption	Wrong ETH data received Wrong ETH data transmitted	P	-		
		ETH _FM_027	ETH communication affected by transmission delay	Data communication: transmission delay	Wrong or missing ETH data received Wrong or missing ETH data transmitted	P	-		ETH_SM_2: Information redundancy techniques on messages, including End to End safing
		ETH _FM_028	ETH communication affected by transmission masquerading	Data communication: transmission masquerading	Wrong ETH message received in multimaster network	P	-	-	
QUADSPI	APB ITF	QSPI _FM_001	Data/address wrong values during QSPI accesses by application software and/or DMA by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to QSPI peripheral by application software and/or DMA	P	-	-	QSPI_SM_0: Periodical read-back of configuration registers
		QSPI _FM_002				T	-	-	QSPI_SM_2: Information redundancy techniques on messages DMA_SM_4: DMA transaction awareness checks
		QSPI _FM_003	Timeout or no arbitration in QSPI APB interface	Bus: No or continuous arbitration	No access to QSPI peripheral by application software or DMA APB bus lock	P	-	-	QSPI_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
QUADSPI	REGISTERS	QSPI _FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No QSPI communication	P	-	-	QSPI_SM_0: Periodical read-back of configuration registers
		QSPI _FM_005		CPU registers, RAM: Change of information caused by soft-errors		T	-	-	
	BAUD GENERATOR	QSPI _FM_006	Wrong or missing clock generation	CPU coding and execution including flag registers: Wrong coding or wrong execution		P	-	-	QSPI_SM_2: Information redundancy techniques on messages
	INTERRUPT CTRL	QSPI _FM_007	Missing interrupt generation on receive/transmit data	CPU coding and execution including flag registers: Wrong coding or wrong execution	No QSPI data processing by application software Missing transmission of QSPI data	P	-	-	QSPI_SM_2: Information redundancy techniques on messages NVIC_SM_1: Expected and unexpected interrupt check by application software
		QSPI _FM_008			No QSPI data processing by application software Missing transmission of QSPI data QSPI receive buffer overrun	T	-	-	-
		QSPI _FM_009	Continuous or unintended QSPI interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software NVIC_SM_1: Expected and unexpected interrupt check by application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
QUADSPI	OUTPUTS	QSPI _FM_010	MISO and/or MOSI signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing QSPI data received Wrong or missing QSPI data transmitted	P	-	QSPI_SM_1: Protocol error signals including hardware CRC	QSPI_SM_2: Information redundancy techniques on messages
QUADSPI	OUTPUTS	QSPI _FM_011	SCLK signals wrong value, open or HiZ	Digital I/O: DC fault model	Missing QSPI data communication	P	-	QSPI_SM_1: Protocol error signals including hardware CRC	QSPI_SM_2: Information redundancy techniques on messages
QUADSPI	OUTPUTS	QSPI _FM_012	SCLK signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing data received/transmitted in multimaster connection	P	-	QSPI_SM_1: Protocol error signals including hardware CRC	QSPI_SM_2: Information redundancy techniques on messages
QUADSPI	CONTROL	QSPI _FM_013	Wrong data processing in receive/transmit processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing QSPI data received	P	-	QSPI_SM_1: Protocol error signals including hardware CRC	QSPI_SM_2: Information redundancy techniques on messages
QUADSPI	CONTROL	QSPI _FM_014	Wrong data processing in receive/transmit processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing QSPI data transmitted	T	-	QSPI_SM_1: Protocol error signals including hardware CRC	QSPI_SM_2: Information redundancy techniques on messages
QUADSPI	CONTROL	QSPI _FM_015	Wrong NSS management	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing data received/transmitted in multimaster network	P	-	-	QSPI_SM_2: Information redundancy techniques on messages
QUADSPI	CONTROL	QSPI _FM_016	Wrong NSS management	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing data received/transmitted in multimaster network	T	-	-	QSPI_SM_2: Information redundancy techniques on messages
QUADSPI	FIFOS	QSPI _FM_017	Single or multiple bit errors in Rx/Tx FIFO cells	CPU registers, RAM: DC fault model for data and addresses	Wrong or missing QSPI data received Wrong or missing QSPI data transmitted	P	-	QSPI_SM_1: Protocol error signals including hardware CRC	-
QUADSPI	FIFOS	QSPI _FM_018	Single or multiple bit errors in Rx/Tx FIFO cells	CPU registers, RAM: Change of information caused by soft-errors	Wrong or missing QSPI data received Wrong or missing QSPI data transmitted	T	-	-	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
QUADSPI	CRC CONTROLLER	QSPI _FM_019	Wrong CRC computation for outgoing message	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong CRC field in outgoing message	P	-	-	-
		QSPI _FM_020				P		-	-
		QSPI _FM_021	Wrong CRC computation for incoming message		CRC mismatch in well-received message	P	-	-	-
		QSPI _FM_022				P	-	-	-
		QSPI _FM_023	CRC error reporting disable		CRC error protection disable	P	-	-	QSPI_SM_2: Information redundancy techniques on messages
		QSPI _FM_024	Unintended CRC error reporting		CRC error propagated so system will switch in safe state	P	-	QSPI_SM_1: Protocol error signals including hardware CRC	-

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
QUADSPI	-	QSPI _FM_025	QSPI communication affected by transmission errors	Data communication: transmission errors	Wrong QSPI data received Wrong QSPI data transmitted	P	-	-	-
		QSPI _FM_026	QSPI communication affected by transmission repetitions	Data communication: transmission repetitions		P	-	-	QSPI_SM_2: Information redundancy techniques on messages
		QSPI _FM_027	QSPI communication affected by transmission deletion	Data communication: transmission deletion	Missing QSPI data received Missing QSPI data transmitted	P	-	-	-
		QSPI _FM_028	QSPI communication affected by transmission resequencing	Data communication: transmission resequencing	QSPI data received in wrong sequence QSPI data transmitted in wrong sequence	P	-	-	-
		QSPI _FM_029	QSPI communication affected by transmission corruption	Data communication: transmission corruption	Wrong QSPI data received Wrong QSPI data transmitted	P	-	-	-
		QSPI _FM_030	QSPI communication affected by transmission delay	Data communication: transmission delay	Wrong or missing QSPI data received Wrong or missing QSPI data transmitted	P	-	-	QSPI_SM_2: Information redundancy techniques on messages
		QSPI_FM _031	QSPI communication affected by transmission masquerading	Data communication: transmission masquerading	Wrong QSPI message received in multimaster network	P	-	-	

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode		Potential effect(s) on the MCU	FM	Control		Recommended action(s)
			Detailed	IEC 61508			Prevention	Detection	
SAI	APB ITF	SAI_FM_001	Data/address wrong values during SAI accesses by application software by APB bus	CPU coding and execution including flag registers: Wrong coding or wrong execution	No or wrong access to SAI peripheral by application software and/or DMA	P	-	-	SAI_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
		SAI_FM_002	Data/address wrong values during SAI accesses by application software and/or DMA by APB bus			T	-	-	DMA_SM_4: DMA transaction awareness checks
		SAI_FM_003		Bus: No or continuous arbitration	No access to SAI peripheral by application software or DMA APB bus lock	P	-	-	SAI_SM_0: Periodical read-back of configuration registers DMA_SM_4: DMA transaction awareness checks
	REGISTERS	SAI_FM_004	Wrong configuration register(s) value	CPU registers, RAM: DC fault model for data and addresses	No SAI communication	P	-	-	SAI_SM_0: Periodical read-back of configuration registers
		SAI_FM_005		CPU registers, RAM: Change of information caused by soft-errors		T	-	-	-
	INTERRUPT CTRL	SAI_FM_006	Missing interrupt generation on receive/transmit data	CPU coding and execution including flag registers: Wrong coding or wrong execution	No SAI data processing by application software	P	-	-	SAI_SM_1: SAI output loopback scheme NVIC_SM_1: Expected and unexpected interrupt check by application software
		SAI_FM_007			No SAI data processing by application software	T	-	-	NVIC_SM_1: Expected and unexpected interrupt check by application software
		SAI_FM_008	Continuous or unintended SAI interrupt request generation	Interrupt: no or continuous interrupt	Application software flow perturbation Application software executed at slower speed	P	-	-	CPU_SM_1: Control flow monitoring in application software running on M4 CPU NVIC_SM_1: Expected and unexpected interrupt check by application software

Table 2. FMEA results - STM32F4 Series (continued)

Module	Submodule	Failure mode code	Potential failure mode: Detailed	Potential failure mode: IEC 61508	Potential effect(s) on the MCU	FM	Control: Prevention	Control: Detection	Recommended action(s)
SAI	OUTPUTS	SAI _FM_009	TX and/or RX signals wrong value, open or HiZ	Digital I/O: DC fault model	Wrong or missing SAI data stream received Wrong or missing SAI data stream transmitted	P	-	-	SAI_SM_2: 1oo2 scheme for SAI module SAI_SM_1: SAI output loopback scheme
	CONTROL	SAI _FM_010	Wrong data processing in receive/transmit processing control units	CPU coding and execution including flag registers: Wrong coding or wrong execution	Wrong or missing SAI data stream received	P	-	-	
		SAI _FM_011			Wrong or missing SAI data stream transmitted	T	-	-	
		SAI _FM_012	Missed SAI wakeup on transmission/receive request		Missing SAI data received Missing SAI data transmitted	P	-	-	
	-	SAI _FM_017	SAI communication affected by transmission errors	Data communication: transmission errors	Wrong SAI data received Wrong SAI data transmitted	P	-	-	
		SAI _FM_018	SAI communication affected by transmission repetitions	Data communication: transmission repetitions		P	-	-	
		SAI _FM_019	SAI communication affected by transmission deletion	Data communication: transmission deletion	Missing SAI data received Missing SAI data transmitted	P	-	-	
		SAI _FM_020	SAI communication affected by transmission resequencing	Data communication: transmission resequencing	Message sent or received in wrong order	P	-	-	
		SAI _FM_021	SAI communication affected by transmission corruption	Data communication: transmission corruption	Wrong SAI data received Wrong SAI data transmitted	P	-	-	
		SAI _FM_022	SAI communication affected by transmission delay	Data communication: transmission delay	Wrong or missing SAI data received Wrong or missing SAI data transmitted	P	-	-	
		SAI _FM_023	SAI communication affected by transmission masquerading	Data communication: transmission masquerading	Unintended message reception	P	-	-	

3 Revision history

Table 3. Document revision history

Date	Revision	Changes
14-Jun-2018	1	Initial release.
28-Jan-2019	2	Updated *Table 2: FMEA results - STM32F4 Series*.
21-Jun-2019	3	Updated functional safety documentation framework. Updated document title. Minor text edits across the whole document.